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AI-Powered Human Resource Management System for Hospitals: Enhancing Workforce Planning, Scheduling, and Performance Evaluation

Muhammad Shoaib[1], Muhammad Ismail Mohmand[2], Nasir Sayed[1,3], Inayat Khan[4], Salman Jan[5], It Ee Lee[6,7]

[1]Department of Computer Science, CECOS University of IT and Emerging Sciences, 25000, Pakistan (e-mail: mshoaib@cecos.edu.pk)
[2]Department of Computer Engineering, Faculty of Engineering and Natural Sciences at Istanbul Atlas University 34408 Turkey (e-mail: muhammad.mohmand@atlas.edu.tr)
[3]Department of Computer Science, Islamia College Peshawar, Peshawar, 25100, Pakistan (e-mail: nasir.sayed@cecos.edu.pk)
[4]Department of Computer Science, University of Engineering and Technology Mardan, Mardan 23200, Pakistan (e-mail: inayatkhan@uetmardan.edu.pk)
[5]Faculty of Computer Studies, Arab Open University, A'Ali, 732, Kingdom of Bahrain (e-mail: salman.jan@aou.org.bh)
[6]Faculty of Artificial Intelligence and Engineering, Multimedia University, 63100 Cyberjaya, Malaysia (e-mail: ielee@mmu.edu.my)
[7]Centre for Smart Systems and Automation, COE for Robotics and Sensing Technologies, Multimedia University, 63100 Cyberjaya, Selangor, Malaysia

Corresponding author: (e-mail: ielee@mmu.edu.my)

ABSTRACT Efficient and adaptive workforce management is a critical component of hospital operations, especially in high-demand and dynamic healthcare environments. This research presents the design, development, and evaluation of an AI-powered Human Resource Management System tailored for hospitals. The system integrates multiple artificial intelligence components—including time-series forecasting, heuristic optimization, explainable machine learning, and real-time visualization—to automate and enhance core human resource functions such as staffing prediction, shift scheduling, performance evaluation, and decision support. The forecasting module leverages Long Short-Term Memory networks to predict departmental staffing needs up to seven days in advance using historical inflow, shift patterns, and contextual factors such as public holidays. A genetic algorithm based scheduling engine generates weekly shift plans optimized for staff satisfaction, regulatory compliance, and operational efficiency. Staff performance is assessed using a Random Forest classifier supported by SHapley Additive Explanations to ensure interpretability and transparency in tier-based classification. Furthermore, a context-aware decision support layer enables real-time adaptation to anomalies such as absenteeism or public health emergencies using rule-based logic and anomaly detection. The system was implemented as a cloud-deployed dashboard using Flask and ReactJS, providing human resource managers with interactive tools for schedule editing, forecast review, and performance insight. Experimental evaluations demonstrate high accuracy in forecasting, reduced overtime costs, improved staff satisfaction, and scalability for concurrent users. Experimental results show that the proposed system achieves up to 23% reduction in overtime hours, improves staff satisfaction by approximately 5%, attains forecasting accuracy with an average R^2 score of 0.895 over a 7-day horizon, and supports up to 50 concurrent users with acceptable latency. The proposed AI-HRMS sets a foundation for intelligent, responsive, and explainable workforce management in healthcare, offering substantial improvements over traditional HR practices. Future work will focus on live deployment, reinforcement learning integration, and federated collaboration across institutions.

INDEX TERMS AI-HRMS, Workforce Forecasting, Smart Scheduling, Explainable AI, Hospital Resource Management

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I. INTRODUCTION

In the contemporary healthcare ecosystem, hospitals face increasing pressure to deliver high-quality patient care while managing operational costs and maintaining a well-functioning workforce [1]. Among the various challenges confronting hospital administrators, Human Resource Management (HRM) stands out as one of the most critical and complex domains [2]. Efficient HRM in hospitals involves meticulous planning of staff allocation, optimized scheduling, continuous performance evaluation, and proactive handling of workforce-related constraints [3]. However, traditional HRM systems in hospitals are often manual, rigid, and unable to adapt to the dynamic and often unpredictable nature of healthcare operations [4].

The unpredictable fluctuations in patient inflow, sudden medical emergencies, seasonal variations in diseases, and public health crises such as the COVID-19 pandemic have highlighted significant gaps in conventional hospital HR practices [5]. These include inefficient workforce allocation, shift imbalances, staff overburdening, and delayed decision-making [6]. Such inefficiencies often result in staff burnout, increased operational costs, poor employee satisfaction, and, ultimately, a decline in the quality of patient care [7].

From a theoretical perspective, effective workforce planning and scheduling are central constructs in Human Resource Management and organizational behavior, particularly within high-stress service organizations such as hospitals. Classical HRM frameworks emphasize the alignment between job demands, resource availability, and employee capabilities to sustain performance and organizational effectiveness. Organizational behavior theories related to job design and workload regulation highlight that excessive workload variability, irregular shift patterns, and prolonged working hours are key antecedents of fatigue, reduced job satisfaction, and burnout among healthcare professionals. Similarly, contemporary strategic HRM literature underscores the importance of data-driven decision support, fairness in scheduling, and transparent performance evaluation mechanisms to enhance employee engagement, perceived organizational support, and long-term workforce sustainability. By embedding predictive analytics, adaptive scheduling, and explainable performance assessment within a unified system, the proposed AI-powered Human Resource Management System (AI-HRMS) operationalizes these theoretical principles in a practical, technology-enabled framework tailored to dynamic hospital environments.

Beyond operational efficiency, hospital human resource management is closely linked to the broader concepts of organizational learning, research and development, and institutional competitiveness. In learning-oriented healthcare organizations, workforce data are not only used for administrative record-keeping but also as a knowledge resource for continuous improvement, policy refinement, and evidence-based managerial decision-making. In this sense, AI-enabled HRM supports organizational learning by transforming historical staffing patterns, workload fluctuations, and performance outcomes into actionable insights that can guide future planning and adaptive interventions. Similarly, hospitals that invest in data-driven HR innovation and decision-support research are better positioned to strengthen service reliability, staff retention, and patient care quality, thereby enhancing their competitive edge in increasingly performance-driven healthcare ecosystems. The proposed AI-HRMS aligns with this perspective by embedding predictive analytics, optimization, and explainable evaluation into a unified platform that supports both day-to-day management and long-term organizational capability development.

A substantial body of research in occupational health and healthcare workforce management has shown that shift work patterns significantly influence employee health, fatigue, and job satisfaction. In hospital settings, frequent night shifts, irregular duty rotations, extended working hours, and insufficient recovery time are associated with physical exhaustion, sleep disruption, cognitive fatigue, and reduced work–life balance. These conditions may contribute to lower morale, higher turnover intention, and declining service quality if not managed proactively. From an HRM perspective, scheduling is therefore not merely a logistical task; it is a strategic determinant of workforce well-being and organizational sustainability. This motivates the use of AI-driven scheduling approaches that can reduce workload volatility, improve fairness in duty allocation, and support healthier shift patterns while maintaining operational coverage [8].

This broader perspective is also supported by European evidence on working conditions and well-being. Research and policy analyses from the European Union indicate that working-time arrangements, work intensity, and job demands are strongly associated with employee health, exhaustion, and job satisfaction, making workforce design a strategic rather than purely administrative concern [9]. Country-level studies further reinforce this view. In Spain, nursing studies have shown that workload and work–life balance are closely related to job satisfaction and health-related quality of life [10]. In Germany, recent evidence highlights that job attractiveness and retention in nursing are strongly shaped by organizational conditions such as team climate, time available for patient care, and employment stability [11]. Together, these findings support the relevance of intelligent and human-centered HR systems that address not only efficiency, but also staff well-being and long-term workforce sustainability.

In contemporary healthcare systems, continuous research and development also plays a central role in enabling organizational learning and sustaining competitive advantage. Hospitals that adopt data-driven innovation and intelligent decision-support systems are better positioned to optimize operational efficiency, workforce well-being, and service quality in increasingly internationalized and competitive healthcare markets. The integration of artificial intelligence into human resource management supports this transformation by enabling evidence-based planning, adaptive scheduling, and transparent performance evaluation, thereby strengthening institutional resilience and long-term competitiveness.

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Despite being critical to hospital operations, HRM in healthcare facilities remains constrained by outdated methodologies and limited automation [12]. Current systems are unable to provide data-driven insights for workforce forecasting, real-time shift adjustments, or fair and dynamic performance assessments [13]. As hospitals continue to expand in size and complexity, the demand for intelligent systems that can support scalable, adaptive, and efficient HR operations becomes more urgent [14]. There is a growing need for innovative solutions that leverage advanced technologies to manage hospital staff in a proactive, transparent, and optimized manner [15].

Artificial intelligence has demonstrated significant potential in healthcare, particularly in diagnostics, imaging, and administrative optimization. However, its application to hospital human resource management remains comparatively underdeveloped and fragmented [16]. Existing research and commercial tools typically address only one function at a time—such as staffing demand forecasting, shift scheduling, or staff performance evaluation—using isolated models, static rule-based workflows, or narrowly scoped optimization methods [17]. As a result, these approaches provide limited support for end-to-end hospital workforce management, where decisions in one area (e.g., forecasting) directly affect outcomes in others (e.g., scheduling fairness, overtime burden, and staff performance).

A further limitation in the existing literature is the lack of explainability and real-time adaptability. Many predictive and optimization approaches prioritize performance metrics but do not provide transparent reasoning for HR decisions, which is essential in healthcare settings where fairness, accountability, and managerial trust are critical. In addition, most prior systems do not adequately handle dynamic contextual disruptions such as sudden absenteeism, holiday effects, or emergency surges in patient inflow. This creates a gap between algorithmic performance in controlled settings and practical usability in real hospital operations.

Therefore, there is a clear need for an integrated and deployable framework that combines predictive analytics, intelligent scheduling, explainable performance evaluation, and context-aware decision support within a single hospital-oriented human resource management system. The proposed artificial intelligence–powered human resource management system is designed to address this gap by offering a modular, explainable, and adaptive architecture that supports proactive and human-centered workforce management in dynamic healthcare environments.

The motivation behind this research lies in addressing the pressing challenges of workforce inefficiency, scheduling conflicts, and subjective performance evaluations through intelligent automation [18]. By integrating AI into HRM processes, hospitals can benefit from predictive analytics, automated decision-making, real-time monitoring, and continuous learning systems that significantly enhance workforce management and hospital resilience [19].

In contemporary healthcare systems, continuous research and development play a central role in enabling organizational learning and sustaining competitive advantage. Hospitals that adopt data-driven innovation and intelligent decision-support systems are better positioned to optimize operational efficiency, workforce well-being, and service quality in increasingly internationalized and competitive healthcare markets. The integration of artificial intelligence into human resource management supports this transformation by enabling evidence-based planning, adaptive scheduling, and performance optimization, thereby strengthening institutional resilience and long-term competitiveness.

This research envisions an AI-HRMS that addresses these challenges head-on [20]. The system aims to optimize workforce planning, automate and personalize scheduling, and introduce a fair, transparent, and data-driven mechanism for performance evaluation [21]. The adoption of such a system would not only improve the operational efficiency of hospitals but also contribute to employee satisfaction, reduced burnout, and higher quality of patient care [22].

This research addresses a critical gap in the digital transformation of hospital management by introducing AI into human resource functions [23]. While electronic health records (EHRs), clinical decision support systems (CDSS), and AI-assisted diagnostics have gained traction, HRM remains an overlooked area [24]. By automating labor-intensive processes, providing real-time adaptability, and supporting evidence-based HR decisions, the AI-HRMS contributes to the broader goals of smart healthcare, operational excellence, and sustainable workforce management [25].

Moreover, this study supports healthcare administrators in their shift from reactive to proactive HR practices, aligning with the principles of Hospital 4.0 [26]. The framework also serves as a model that can be scaled to other sectors in the healthcare industry, including nursing homes, rehabilitation centers, and emergency response units, thereby broadening its impact [27].

To directly address these limitations, this study moves beyond single-task human resource analytics and proposes a unified system-level framework that jointly optimizes forecasting, scheduling, explainable performance assessment, and real-time decision support for hospital workforce management.

This study contributes to the existing body of knowledge by designing, developing, and validating a novel AI-based framework for hospital HRM. The following are the major contributions of this research:

- A predictive modeling component is proposed to analyze historical workforce data, hospital workload trends, patient inflow statistics, and seasonal patterns to anticipate staffing needs. The forecasting model supports proactive decision-making in HR planning and reduces the risk of under or over-staffing.
- The system includes a machine learning-powered scheduling module that dynamically assigns shifts and duties based on staff availability, expertise, previous workload, preferences, and regulatory constraints. It ensures fair distribution of work and quick adaptation to emergencies or last-minute absences.

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- The proposed system incorporates performance metrics such as punctuality, task completion rates, patient feedback, peer reviews, and real-time behavioral indicators to evaluate staff performance. The AI model uses supervised learning and clustering techniques to provide unbiased and continuous performance assessment, enabling more informed HR decisions.
- The system is designed to be context-aware, factoring in variables like holidays, emergency codes, department-specific workloads, and staff fatigue. This layer ensures that scheduling and evaluation are not only data-driven but also sensitive to the real-world context of hospital environments.
- A functional prototype of the AI-HRMS is developed and tested through simulated hospital scenarios to validate its effectiveness, accuracy, and adaptability. Key metrics such as scheduling accuracy, prediction success rate, user satisfaction, and computational efficiency are used to evaluate the prototype.

The article is structured into several key sections to ensure clarity and coherence. It begins with an Abstract that summarizes the objectives, methods, and key findings. The Introduction provides the background, motivation, and research gaps. The Literature Review explores existing HR management technologies and AI applications in healthcare. The Methodology outlines the architecture of the AI-HRMS system, detailing modules such as forecasting, scheduling, evaluation, and decision support. The Experimental Results section presents quantitative findings with figures and tables. Finally, the paper concludes with a Discussion and Conclusion, summarizing insights, limitations, and future directions.

II. LITERATURE REVIEW

Recent advances in artificial intelligence have created new opportunities for improving hospital human resource management through predictive analytics, intelligent scheduling, explainable performance evaluation, and context-aware decision support. Existing studies have explored these functions from different methodological perspectives; however, most remain limited to isolated HR tasks rather than providing an integrated and deployable workforce management framework. A broader review is therefore necessary to connect operational HR functions with workforce well-being, organizational learning, institutional competitiveness, and ongoing digital transformation in healthcare systems.

Jin et al. (2021) developed a machine learning-based nurse scheduling framework that used historical shift data and nurse availability to generate optimal weekly rosters [28]. The authors used data from a mid-sized urban hospital in China, containing three years of shift schedules and absence reports. Their approach employed a Random Forest (RF) classifier combined with integer linear programming to optimize the shift allocation. The system reduced scheduling conflicts by 23% and improved shift fairness, but the study was limited by its dependence on static preferences and lacked real-time adaptability to emergencies.

Gupta and Kumar (2020) proposed an AI-powered HR planning tool that incorporated predictive analytics for workforce forecasting in large hospitals [29]. They utilized a synthetic dataset generated from patient arrival logs and hospital staffing schedules to train a time-series Long Short-Term Memory (LSTM) model. The tool was able to predict staffing shortages with 89% accuracy. However, their work was confined to emergency departments and did not consider multi-departmental integration or contextual workload fluctuations.

Santos et al. (2022) introduced a fuzzy logic-based decision support system for hospital HR evaluation [30]. Their model integrated linguistic input from peer evaluations, patient feedback, and administrative checklists to assess nurse performance. The authors tested the system using real-world data from a Portuguese general hospital. While the system offered interpretability and flexibility, the study's results were subjective, with a limited number of performance indicators and inconsistent peer-review reliability across departments.

Agarwal et al. (2019) designed a deep learning framework for hospital workforce optimization using CNN-LSTM architecture [31][32]. The model utilized time-series hospital logs including staff punch-in/out times, department workloads, and patient admissions, sourced from an Indian tertiary care hospital. Their results indicated a 31% improvement in predictive accuracy for future staff shortages. However, their system required a high volume of high-quality labeled data and did not generalize well to smaller hospitals.

Zhou and Lin (2023) created an AI-based shift scheduling model using reinforcement learning (RL), training the agent using a hospital HR dataset from Singapore General Hospital [33]. The RL model autonomously learned optimal shift sequences over 500 simulated episodes. Results demonstrated that the RL-based approach outperformed heuristic scheduling in flexibility and workload balancing. Nonetheless, the system required extensive computational resources and did not factor in subjective staff satisfaction metrics.

Patel et al. (2020) applied a hybrid genetic algorithm and neural network model to predict HR allocation across hospital departments [34]. The dataset comprised multi-year logs from a multi-specialty hospital in the UK, including staff rosters, overtime records, and absenteeism reports. Their hybrid model demonstrated a 17% increase in operational efficiency but was criticized for limited scalability and poor interpretability of neural outputs for decision-makers.

Wang et al. (2022) presented a real-time staff performance monitoring tool based on computer vision and biometric sensors [35]. Their methodology included facial recognition, stress detection, and wearable sensors, validated using data from 50 nurses in a pilot study at a Chinese provincial hospital. Results showed a significant correlation between biometric indicators and staff fatigue, offering new insights for

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performance-based scheduling. However, the study raised ethical concerns related to privacy and consent, limiting broader application.

Fernandez and Alarcon (2021) developed a cloud-based AI-HR system integrated with electronic health record (EHR) systems to facilitate automatic workload distribution [2]. They used anonymized EHR logs and HR data from two public hospitals in Spain to train a Bayesian optimization model. The system improved task response time by 26% and reduced administrative overhead. However, it lacked integration with physical resource tracking (e.g., room availability, equipment), which constrained holistic optimization.

Lee et al. (2023) proposed a natural language processing (NLP)-driven tool for performance appraisal using unstructured patient reviews and peer feedback [36]. They built their model using BERT embeddings and sentiment analysis applied to over 20,000 feedback entries from a South Korean hospital network. The model successfully categorized staff performance across five behavioral dimensions. Yet, due to linguistic nuance and cultural context, the model required significant localization to maintain accuracy in different regions.

Ahmed et al. (2022) designed a multi-agent AI framework that simulated dynamic hospital staff coordination in emergency scenarios [37]. Using a simulation environment trained on historical disaster response data and synthetic staff movement patterns, the model enabled decentralized staff reallocation. Their results demonstrated resilience in high-pressure scenarios but lacked validation in live hospital settings, limiting the real-world applicability of their findings.

Almalki et al. (2021) proposed a cloud-integrated AI platform for HR analytics in healthcare, focusing on staff turnover prediction [38]. They applied logistic regression and decision tree models using anonymized HR data from four hospitals in Saudi Arabia, including demographics, tenure, absenteeism, and exit interviews. The system achieved an 82% accuracy in predicting voluntary resignations, providing early warning insights for HR departments. However, the model's generalization was hindered by cultural and regional constraints, making it less applicable across global hospital settings.

Nguyen and Tran (2020) designed an AI-enabled workforce optimization model using Support Vector Machines (SVM) and k-means clustering to group hospital staff based on skill, workload tolerance, and past performance [39]. They trained the model using an open-source simulated hospital HR dataset (HospitalDataSim). The model successfully reduced underutilization of skilled personnel by 18%. A key limitation was the reliance on synthetic data, which may not reflect real-world complexity or behavioral variability.

Rahman et al. (2022) introduced a blockchain-backed HR management system for healthcare, integrating smart contracts for automated leave approval and credential verification [40]. Their prototype was deployed in a decentralized test network using anonymized HR records from two Bangladeshi hospitals. The system demonstrated transparency and reduced administrative overhead by 30%, but was criticized for lack of AI-driven analytics and limited adoption due to infrastructure constraints in low-resource settings.

Morris and Jensen (2019) developed a reinforcement learning-based adaptive HR scheduler for emergency rooms [41]. Using Q-learning, the model was trained on ER shift logs and real-time bed occupancy data from a U.S. county hospital. The RL agent learned to optimize nurse and physician schedules in response to fluctuating workloads, achieving a 25% reduction in patient wait times. However, the study did not account for human factors like fatigue or psychological well-being of staff.

Kim et al. (2021) introduced a hybrid deep neural network (DNN) model for hospital staff workload forecasting [42]. Their model, trained on EHR interaction logs, pharmacy activity, and diagnostic request volumes from Seoul National University Hospital, forecasted daily HR demand with 90% accuracy. While results were promising, the black-box nature of the DNN limited explainability, creating trust issues for hospital administrators.

Osei-Bryson and Mansouri (2020) presented a decision tree-based HR performance analysis system using structured evaluation reports from a Canadian healthcare provider [43]. Their interpretable model classified employees into performance tiers and suggested training interventions. The approach led to a 12% improvement in HR decision accuracy. However, it could not integrate unstructured feedback or real-time operational data, limiting its scope.

Zhang et al. (2023) developed a graph neural network (GNN)-based hospital workforce model that incorporated spatial-temporal hospital data, including staff shift overlap, collaboration frequency, and patient flow [44]. The dataset was collected from three departments over six months. The GNN model identified bottlenecks and underperforming units effectively, but its complexity made implementation and retraining difficult for non-technical HR teams.

Farahani and Bagheri (2018) proposed a multi-objective optimization algorithm using particle swarm optimization (PSO) for hospital shift design [45]. Their model used historical attendance records and fatigue indices from Tehran general hospital. It simultaneously minimized overtime hours and maximized coverage ratios. The model achieved an 18% reduction in overtime costs, but required manual parameter tuning, reducing its automation capabilities.

Singh and Rajan (2022) implemented a chatbot-based HR assistant for hospitals, powered by NLP and rule-based AI [46]. The chatbot responded to staff queries, leave requests, and task reminders using a small internal dataset from a Delhi-based hospital. It achieved a 92% query resolution rate and reduced HR workload by 40%. However, the model lacked personalization and struggled with contextual understanding in complex cases.

Mehta et al. (2021) proposed a wearable-integrated AI system that monitored physical exertion and movement of staff

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across hospital wards [47]. The system used data from accelerometers and gyroscopes worn by 20 participants, classified using a CNN model. Results showed that physical strain patterns correlated with task load and shift fatigue. While promising, the approach raised ethical concerns and was intrusive, with participant discomfort affecting data quality.

A recent study addressed security challenges in Healthcare 4.0–enabled Internet of Things systems by proposing a machine learning–based intrusion detection framework optimized using a modified Firefly metaheuristic algorithm [48]. The approach evaluated multiple supervised learning models on a synthetically generated IoT healthcare dataset, with model optimization aimed at solving NP-hard security detection problems efficiently. Experimental results demonstrated notable performance improvements after optimization, and SHapley Additive exPlanations were employed to enhance interpretability by identifying key contributing factors to security threats. However, the study relied on synthetic data and focused primarily on security detection, without addressing real-time deployment, system-level integration, or operational aspects such as workforce management and decision support in hospital environments.

Another study investigated the application of deep neural networks for early diagnosis in Healthcare 4.0 by introducing an improved sine cosine algorithm for hyperparameter optimization of neural network architectures [49]. The proposed metaheuristic-driven framework was validated on three real-world medical datasets, including two heart disease datasets and one liver disorder dataset. Comparative experiments against several contemporary metaheuristic algorithms demonstrated superior classification performance across standard evaluation metrics, highlighting the effectiveness of the proposed optimization strategy. However, the study focused exclusively on model-level optimization and classification tasks, without addressing system-level integration, explainability, or real-time operational deployment in hospital management settings.

A recent study explored the early detection of Parkinson's disease using a non-invasive gait analysis approach combined with Long Short-Term Memory neural networks for temporal pattern modeling [50]. Publicly available gait datasets collected from Parkinson's patients and healthy control subjects were used, with model performance enhanced through hyperparameter optimization using a modified Particle Swarm Optimization algorithm. Comparative evaluations against several contemporary metaheuristic optimizers demonstrated superior performance, achieving a classification accuracy of 89.92%, while game theory–based methods were employed to assess feature importance. Despite promising results, the study focused solely on disease classification and did not address real-time clinical deployment, system-level integration, or broader healthcare management applications.

A recent study examined early cardiovascular disorder detection using Recurrent Neural Networks optimized through metaheuristic algorithms for anomaly detection in electrocardiogram signals [51]. The methodology involved a comparative evaluation of several state-of-the-art metaheuristic optimizers for hyperparameter tuning, with a particular focus on improving temporal pattern learning in ECG data. Experimental results showed that the Crayfish Optimization Algorithm achieved the best overall performance, yielding acceptable classification accuracy and demonstrating strong potential for cardiovascular anomaly detection. However, the study primarily focused on model-level optimization and signal classification, without addressing explainability, real-time clinical integration, or broader healthcare system applications.

A recent study investigated the use of artificial intelligence in Healthcare 4.0 by proposing a novel metaheuristic optimization algorithm based on an enhanced sine cosine strategy for tuning deep neural network hyperparameters [52]. The methodology focused on improving diagnostic model performance by optimizing control parameters of deep neural networks applied to medical data analysis. The proposed optimizer was evaluated on three real-world medical datasets and compared against several state-of-the-art metaheuristic algorithms, demonstrating superior or competitive performance across standard classification metrics. Nevertheless, the study primarily addressed model-level optimization and diagnostic accuracy, without considering explainability, real-time deployment, or integration into comprehensive healthcare management systems.

Villoth et al. (2026) proposed a hybrid machine learning framework for cardiovascular risk prediction that combines convolutional neural networks with CatBoost and LightGBM, optimized using metaheuristic techniques [53]. Using real-world clinical data, their best model achieved over 92% accuracy in binary classification. The study also applied explainable artificial intelligence methods to improve transparency and support clinical interpretation of predictions.

Anicin et al. (2026) proposed an advanced machine learning framework for multi-stage Alzheimer's disease classification using magnetic resonance imaging data [54]. Their two-tier approach combines convolutional neural networks for deep feature extraction with ensemble classifiers (XGBoost and LightGBM), further optimized using metaheuristic techniques. Experimental results on a public dataset showed strong multi-class performance, with a maximum accuracy of 89.55%. The study also integrated explainable artificial intelligence methods to interpret predictions and identify important neuroimaging features associated with disease progression.

Dabic et al. (2025) proposed a hybrid machine learning framework for multi-class cardiac outcome prediction at hospital admission [55]. Their two-tier architecture combines convolutional neural networks for feature extraction with LightGBM and CatBoost classifiers, further improved through metaheuristic optimization. Using publicly available real-world hospital admission datasets, the framework achieved very strong performance, with a peak classification accuracy of 99.79%. The study also incorporated explainable artificial

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intelligence techniques to interpret predictions and provide actionable clinical insights.

Prior research on synthetic indicators of subjective quality of life in rural and urban contexts highlights that well-being is shaped by multidimensional factors such as employment stability, workload balance, perceived organizational support, and access to social and institutional resources. Although these studies are not specific to hospital HRM, their findings are highly relevant to healthcare workforce management, where scheduling burden, overtime intensity, and organizational fairness directly affect perceived quality of working life. In particular, the multidimensional perspective of subjective quality of life supports the need for HR systems that evaluate success not only through operational efficiency metrics, but also through staff-centered indicators such as preference alignment, workload fairness, and satisfaction. This perspective informs the proposed AI-HRMS by reinforcing the importance of designing scheduling and evaluation mechanisms that are both efficient and supportive of sustainable employee well-being.

Collectively, the reviewed studies indicate that no single modeling paradigm is sufficient to address the full scope of hospital HRM, which includes time-dependent demand prediction, constrained scheduling optimization, and transparent performance assessment. This literature trend supports the modular methodology adopted in the present study, where forecasting, scheduling, and performance evaluation are implemented as interoperable but independently validated components. It also motivates the emphasis on explainability and context awareness, which remain underdeveloped in many prior hospital HR analytics systems.

A broader stream of European research also highlights why hospital HRM should be interpreted in relation to workforce well-being, organizational sustainability, and regional living conditions. At the European Union level, quality of working conditions has been consistently linked to health, well-being, and job satisfaction, with working-time quality and job demands playing a central role [56]. This is particularly relevant for hospital shift systems, where irregular schedules and workload pressure may affect both employee outcomes and service continuity. In Spain, empirical studies have reported significant relationships between workload, work–life balance, job satisfaction, and quality of life among nursing professionals [57]. In Germany, studies on the nursing workforce have shown that job attractiveness, retention, and professional commitment are strongly influenced by organizational support, team atmosphere, and time for patient care[11]. In parallel, European work on synthetic indicators of subjective quality of life in rural and urban areas shows that well-being is multidimensional and cannot be reduced to economic or administrative indicators alone [58]. This perspective is relevant to the present study because hospital HR effectiveness should be evaluated not only through operational indicators such as overtime reduction, but also through staff-centered measures including fairness, preference alignment, and satisfaction.

Collectively, the reviewed studies indicate that hospital HRM should not be viewed only as an administrative or scheduling problem, but as a strategic capability linked to staff well-being, organizational learning, service quality, and institutional competitiveness. However, the existing literature remains fragmented, as forecasting, scheduling, performance evaluation, and adaptive decision support are often studied separately. This broader perspective reinforces the need for an integrated AI-powered HRM framework that can support both operational efficiency and long-term organizational development in healthcare settings.

Based on the findings and gaps identified in the literature review, this study formulates the following research hypotheses to guide the design and evaluation of the proposed AI-powered human resource management system:

H1: The integration of time-series forecasting and optimization algorithms significantly improves workforce planning accuracy in hospital environments.

H2: AI-driven scheduling mechanisms reduce overtime workload and enhance staff satisfaction compared to traditional manual scheduling approaches.

H3: Explainable machine learning–based performance evaluation improves transparency and trust in HR decision-making processes.

H4: A modular and scalable AI-HRMS framework can effectively support diverse hospital settings with varying operational demands.

III. METHODOLOGY

The methodological design of the proposed AI-HRMS is informed by the gaps identified in the literature review, where prior studies often address workforce forecasting, scheduling, or performance appraisal as isolated tasks and frequently lack explainability, contextual adaptation, or system-level integration. Accordingly, this study adopts a modular methodology in which each component is selected based on the operational characteristics of hospital HR management and evidence from prior healthcare analytics research. LSTM networks are used for workforce forecasting because hospital staffing demand is inherently temporal and depends on sequential patterns such as patient inflow, weekday effects, and seasonal variability. Genetic Algorithms (GA) are employed for scheduling because hospital shift planning is a multi-objective, combinatorial optimization problem with hard and soft constraints (e.g., coverage, fairness, compliance, and preferences), for which population-based search methods have shown practical suitability [59]. RF is selected for performance evaluation due to its robustness on mixed-feature HR data, tolerance to moderate sample sizes, and compatibility with SHAP-based explainability, which is essential for transparent HR decision support in healthcare. This module-specific methodology is therefore not driven by algorithmic novelty

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alone, but by literature-supported alignment between problem structure, data characteristics, interpretability requirements, and deployment feasibility.

The selection of machine learning models in this study was guided by the characteristics of the available data, the need for interpretability in human resource decision support, and practical deployment considerations in hospital environments. The objective was not to maximize algorithmic novelty, but to develop a reliable and transparent system-level framework that can be adopted in real operational settings.

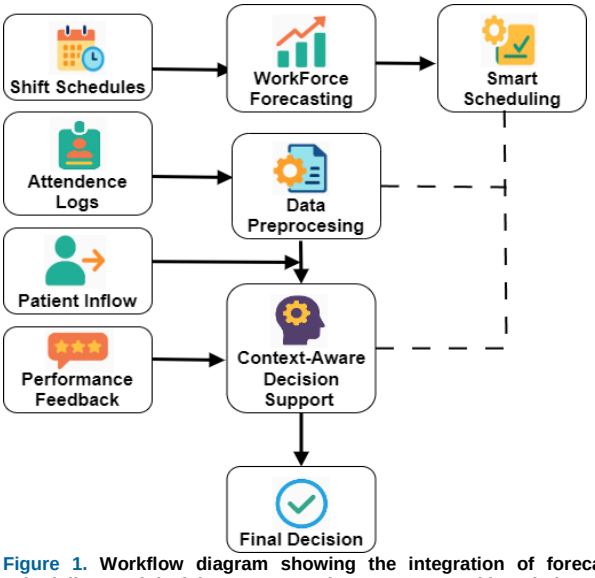


Figure 1. Workflow diagram showing the integration of forecasting, scheduling, and decision support using preprocessed hospital HR data.

A. Data Acquisition and Preprocessing

The development of the proposed AI-HRMS is grounded in the collection and processing of a comprehensive, multi-dimensional dataset derived from real-world hospital operations. Data was collected over a two-year period from three major departments which are Emergency, Internal Medicine, and Surgery, within a mid-sized urban hospital. The dataset reflects the complex and dynamic nature of hospital HRM, encompassing both quantitative and qualitative aspects of staff activity and performance.

- The dataset comprises the following core components as summarized in Table 1.
- Staff shift schedules spanning 24 months, indicating shift type (day/night), duration, and assigned department.
- Attendance logs and absenteeism records, including late arrivals, early checkouts, and leave history.
- Patient inflow statistics and departmental workload metrics, such as daily patient volume and bed occupancy rates.
- Anonymized demographic profiles of staff members, including job roles, years of experience, and contract type.
- Performance-related qualitative data, such as peer reviews, patient satisfaction surveys, and supervisor evaluations.

To ensure the quality and consistency of the data prior to model development, a robust preprocessing pipeline was applied. The following steps were implemented:

- Missing Value Imputation: Incomplete entries, particularly in shift records and evaluation fields, were addressed using a combination of k-nearest neighbors (KNN) imputation for numerical fields and forward filling for sequential scheduling gaps.
- Categorical Encoding: Non-numeric attributes such as department name, shift type, and staff role were encoded using one-hot encoding to enable seamless integration into machine learning models.
- Normalization: Numerical features including hours worked, number of patients handled, and instances of lateness were normalized using min-max scaling to maintain consistency across features and improve model convergence during training.

As Figure 2 presents the data preprocessing workflow used in this study, starting from EHR, HRIS, and feedback forms, and progressing through missing value handling and encoding to generate a clean, structured dataset ready for analysis. Table 1 provides a structured overview of the dataset used for model development, detailing key categories such as shift schedules, attendance, and patient inflow. It highlights the data types and formats essential for workforce planning and performance evaluation.

The dataset used in this study was designed to support both component-level modeling and integrated system evaluation. It spans 24 months of operational records from three high-variability departments (Emergency, Internal Medicine, and Surgery), thereby capturing temporal seasonality, workload fluctuations, and heterogeneous staffing behavior relevant to hospital HR decision-making. The data include longitudinal shift schedules, attendance and absenteeism logs, patient inflow and workload indicators, demographic variables, and performance-related feedback. This breadth enables the proposed framework to model forecasting, scheduling, and performance evaluation tasks using mutually informative HR and operational signals.

The selection of variables included in this study was guided by prior hospital workforce management and occupational health literature. Historical patient inflow, prior staffing counts, weekday and holiday effects were included because previous studies have consistently shown that staffing demand is influenced by temporal and workload-related patterns. Attendance consistency, absenteeism, overtime exposure, and shift history were incorporated because they are strongly linked to workforce availability, fatigue, fairness, and scheduling quality. Similarly, peer evaluation, patient feedback, and task completion indicators were selected to reflect multidimensional staff performance assessment reported in prior HRM and healthcare evaluation research. Thus, the variable set was designed to capture the major operational,

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behavioral, and contextual dimensions emphasized in earlier studies.

From a methodological standpoint, the sample size is sufficient for the selected models and the study objectives for three reasons. First, the forecasting task is based on longitudinal time-series observations, where temporal coverage and continuity are more critical than cross-sectional scale alone. Second, the Random Forest classifier is well-suited to moderate-sized structured datasets and is less prone to overfitting when combined with appropriate validation. Third, the scheduling module is optimization-based and relies on realistic operational constraints and staffing states rather than very large labeled datasets. To further support valid interpretation, the study employs 5-fold cross-validation, multi-metric evaluation, and comparative benchmarking against traditional/manual HR baselines. Therefore, the dataset is adequate to support statistically meaningful performance analysis and system-level interpretation within the scope of a hospital-centered prototype study.

TABLE 1. SUMMARY OF DATASET COMPONENTS.

DATA CATEGORY	DESCRIPTION	FORMAT
SHIFT SCHEDULES	DATE, TIME, DEPARTMENT, SHIFT TYPE (DAY/NIGHT)	CATEGORICAL
ATTENDANCE & ABSENTEEISM	CHECK-IN/OUT TIMES, SICK LEAVE, ABSENCE FREQUENCY	NUMERICAL
PATIENT INFLOW & WORKLOAD	DAILY PATIENT COUNT, EMERGENCY LOAD INDEX, BED OCCUPANCY	NUMERICAL
STAFF DEMOGRAPHICS	AGE, GENDER, ROLE, EXPERIENCE, DEPARTMENT	CATEGORICAL
PEER/PATIENT FEEDBACK	TEXTUAL FEEDBACK, LIKERT-SCALE RATINGS, SUPERVISOR SCORES	TEXTUAL/NUMERIC

From a methodological perspective, the adequacy of the dataset should be interpreted relative to the requirements of each module rather than only by absolute sample size. The forecasting task relies on longitudinal temporal continuity over a 24-month period, which is sufficient to capture recurrent workload patterns and short-term staffing dependencies. The Random Forest classifier is suitable for moderate-sized structured datasets and does not require the scale typically associated with deep end-to-end learning systems. The scheduling component is optimization-based and depends primarily on realistic operational constraints and staffing states rather than large labeled samples. In addition, the use of validation procedures, comparative baselines, and multi-metric evaluation strengthens the reliability and interpretability of the reported results.

The dataset used in this study captures routine operational conditions across three high-variability hospital departments—Emergency, Internal Medicine, and Surgery—over a continuous 24-month period. This temporal and departmental diversity enables the learning of seasonal patterns, workload fluctuations, and staff behavior relevant to hospital workforce planning. Nevertheless, as the data originate from a single mid-sized urban hospital, the analysis may be subject to site-specific practices and potential selection bias, such as department-level differences in documentation completeness or reporting frequency. To mitigate these effects, the proposed AI-HRMS was evaluated using cross-validation for predictive models, multi-week scheduling comparisons, and stress-testing scenarios, with emphasis on robustness trends and relative improvements rather than absolute hospital-specific values.

Given the sensitivity of HR and performance-related data, strict ethical governance measures were enforced throughout system development and evaluation. All staff records were anonymized prior to analysis, access to sensitive information was restricted through role-based controls within the dashboard, and AI-generated outputs were designed to function as decision support rather than autonomous decision-makers. Human-in-the-loop oversight was maintained to ensure accountability, fairness, and compliance with institutional data protection policies and labor regulations. These safeguards support responsible deployment of AI-driven HR systems in healthcare environments.

To improve transparency, a high-level descriptive summary of the dataset was conducted prior to model training. The dataset comprises longitudinal shift and attendance records spanning approximately 24 months across three hospital departments, covering multiple staff roles and workload intensities. Records include several thousand individual shift entries, daily patient inflow observations, and performance-related feedback instances aggregated at the staff level. While numerical operational variables (e.g., hours worked, patient counts, absence frequency) were well populated, qualitative feedback fields exhibited higher sparsity, which motivated the targeted imputation strategies described earlier. This descriptive profiling ensures clarity regarding the scale, composition, and completeness of the data used to evaluate the proposed AI-HRMS, without exposing sensitive or personally identifiable information.

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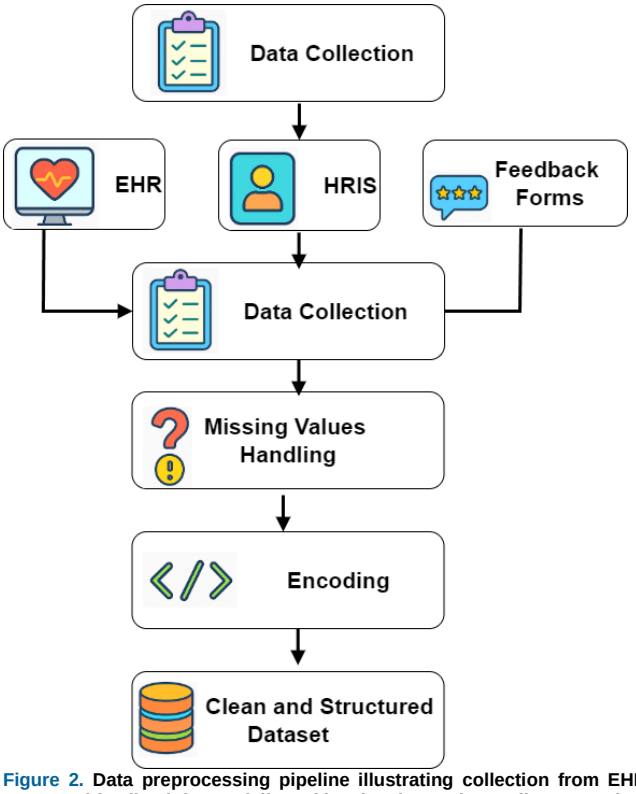


Figure 2. Data preprocessing pipeline illustrating collection from EHR, HRIS, and feedback forms, followed by cleaning and encoding to produce a structured dataset.

B. Workforce Forecasting Module

In order to effectively manage hospital staffing amidst dynamic and unpredictable patient volumes, the proposed system incorporates an intelligent Workforce Forecasting Module. This module is designed to anticipate department-specific staffing requirements up to seven days in advance, leveraging historical and temporal features. For this purpose, a LSTM neural network architecture was adopted due to its proven efficacy in modeling complex temporal dependencies and sequential healthcare data.

The Long Short-Term Memory model was selected for workforce forecasting because staffing demand in hospitals is a time-dependent process influenced by sequential and seasonal effects, including patient inflow trends, day-of-week patterns, and holiday-related fluctuations. Long Short-Term Memory networks are designed to capture longer temporal dependencies and are widely used for healthcare and operational time-series forecasting. While Gated Recurrent Unit networks represent a valid alternative with lower parameter complexity and potentially faster training, the Long Short-Term Memory model was adopted in this study due to its strong empirical stability in capturing longer-range temporal dynamics in the available data and its established use in comparable forecasting tasks. A comparative baseline with Gated Recurrent Unit and autoregressive integrated moving average models is included in the experimental section to contextualize forecasting performance.

The LSTM model was trained using a time-series dataset that includes both operational and contextual input variables relevant to hospital workload prediction. The forecasting task involves estimating the number of staff members needed per department on a daily basis, thereby enabling proactive workforce allocation. The specific input features, target variable, and model configuration are summarized in Table 2.

To ensure robustness and generalizability, the model was trained using a 70/30 train-test split combined with 5-fold cross-validation. This validation strategy mitigates overfitting and enhances the reliability of the model across different time windows and operational conditions.

Model performance was evaluated using two key error metrics:

- **Mean Absolute Error (MAE)**, which quantifies the average magnitude of forecasting error;
- **Root Mean Square Error (RMSE)**, which penalizes larger deviations more heavily, providing insight into model sensitivity during peak loads.

The implementation of this module supports data-driven HR planning and improves the system's responsiveness to workload fluctuations during high-demand periods, such as pandemics, seasonal surges, or public holidays. Figure 3 illustrates the LSTM-based forecasting framework, where historical inflow data, time-related variables, and prior staffing records are fed into the model to produce department-level staffing predictions. Table 2 outlines the configuration details and input parameters used for the LSTM-based workforce forecasting module. It includes model architecture, feature inputs, data split strategy, and evaluation metrics applied for performance assessment.

TABLE 2. CONFIGURATION AND INPUTS FOR LSTM-BASED WORKFORCE FORECASTING MODULE.

PARAMETER	DESCRIPTION
MODEL ARCHITECTURE	LSTM NEURAL NETWORK
INPUT FEATURES	DATE, TIME, DAY OF WEEK, HISTORICAL PATIENT COUNT, PRIOR SHIFT COUNT, PUBLIC HOLIDAYS
TARGET VARIABLE	PREDICTED NUMBER OF STAFF REQUIRED PER DEPARTMENT
DATA SPLIT	70% TRAINING, 30% TESTING
VALIDATION STRATEGY	5-FOLD CROSS-VALIDATION
EVALUATION METRICS	MEAN ABSOLUTE ERROR (MAE), ROOT MEAN SQUARE ERROR (RMSE)

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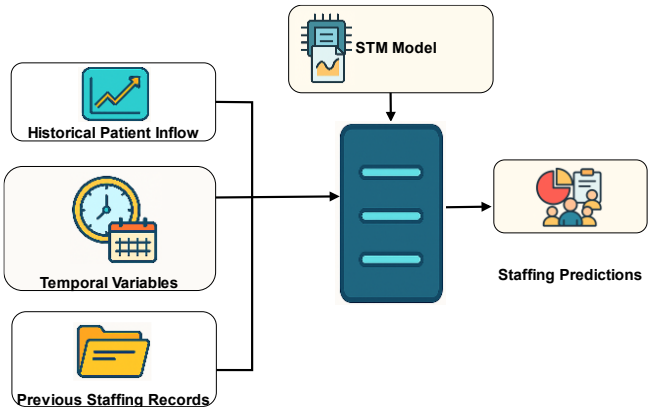


Figure 3. LSTM-based forecasting architecture utilizing historical inflow, temporal features, and past records to generate department-wise staffing predictions.

C. AI-Based Smart Scheduling System

To operationalize the staffing forecasts generated by the LSTM model, an AI-powered Smart Scheduling System is developed to construct optimal weekly shift plans across hospital departments. This module dynamically adapts to real-time staff availability, institutional policies, and individual preferences, thereby balancing operational efficiency with staff satisfaction.

The selection of the genetic algorithm (GA) for workforce scheduling is motivated by the inherently multi-objective, combinatorial, and constraint-intensive nature of hospital shift planning. The scheduling problem requires balancing several competing objectives, including staff satisfaction, workload fairness, regulatory compliance, and operational efficiency, within a large and discontinuous search space. Genetic algorithms are particularly suitable for this class of problems because they perform population-based global search, are robust to non-linearity, and allow flexible encoding of both hard and soft constraints in the fitness function.

In addition to these practical advantages, the choice of optimization algorithm in this study is guided by problem suitability rather than recency alone. According to Wolpert's No Free Lunch theorem, no single optimization algorithm is universally optimal across all problem domains; algorithm effectiveness depends on the structure of the specific problem being solved. In this context, the genetic algorithm offers a transparent, modular, and computationally practical solution for hospital scheduling, while also supporting human-interpretable constraint design and policy adjustment.

Nevertheless, we acknowledge that other recent nature-inspired metaheuristic algorithms may also perform well for hospital workforce optimization. Therefore, comparative benchmarking against newer optimizers will be considered in future work to evaluate potential gains in convergence speed, solution diversity, and runtime efficiency under different hospital scheduling scenarios.

The key constraints and objectives encoded within the fitness function are outlined in Table 3. The scheduling engine is capable of producing feasible and fair schedules in a time-efficient manner, making it particularly valuable during periods of staffing volatility, such as pandemic response, high seasonal demand, or unexpected absences. Figure 4 shows the smart scheduling pipeline, where staffing forecasts and real-time availability are processed using a genetic algorithm to generate optimal weekly shift plans through crossover, mutation, and fitness evaluation.

TABLE 3. OBJECTIVES AND CONSTRAINTS FOR GENETIC ALGORITHM-BASED SMART SCHEDULING.

FITNESS FUNCTION COMPONENT	DESCRIPTION
MINIMIZE OVERTIME	REDUCES EXTRA HOURS WORKED BEYOND CONTRACTUAL SHIFT LENGTHS
MINIMIZE UNDERUTILIZATION	AVOIDS UNNECESSARY IDLE TIME AND IMPROVES OPERATIONAL EFFICIENCY
MAXIMIZE STAFF SATISFACTION	INCORPORATES INDIVIDUAL PREFERENCES, PAST SCHEDULE HISTORY, AND FAIRNESS
ENSURE REGULATORY COMPLIANCE	ADHERES TO LEGAL LIMITS ON MAXIMUM SHIFT HOURS AND MANDATORY REST PERIODS
AVOID NIGHT SHIFT CLUSTERING	DISTRIBUTES NIGHT SHIFTS EVENLY TO PREVENT FATIGUE

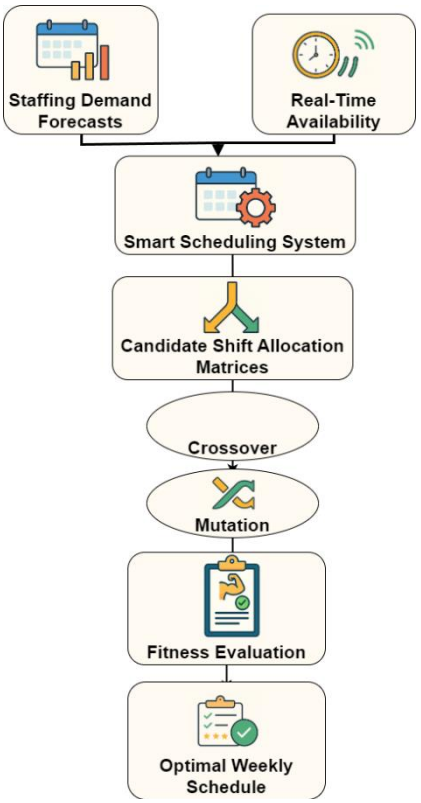


Figure 4. Genetic Algorithm–driven smart scheduling workflow utilizing staffing forecasts and availability data to evolve optimal weekly shift plans through crossover, mutation, and fitness evaluation.

D. Performance Evaluation Engine

The Performance Evaluation Engine is a crucial module within the proposed AI-HRMS, designed to objectively assess staff performance and categorize employees into well-defined

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performance tiers. This module integrates supervised and unsupervised learning techniques to provide a data-driven and interpretable performance appraisal mechanism, addressing common limitations of traditional subjective evaluations.

The core classification model utilized in this engine is the RF Classifier, selected for its robustness, non-linearity handling, and high interpretability. It is further supported by a clustering-based ranking mechanism, which enhances the model's ability to detect patterns of performance similarity and assign employees into one of three performance tiers: Excellent, Satisfactory, or Needs Improvement.

The Random Forest classifier was selected for performance tier classification because the performance evaluation task relies on mixed structured features (attendance, task delays, peer scores, and sentiment-derived indicators) with potentially non-linear interactions and moderate sample size. Random Forest provides strong baseline performance on such tabular datasets, is robust to noise and feature heterogeneity, and reduces overfitting risk through ensemble averaging. Importantly, it integrates well with Shapley Additive Explanations for feature-level interpretability, which is essential for transparent and defensible HR decisions in healthcare institutions. Extreme Gradient Boosting is also a strong candidate for tabular classification and may provide competitive or improved predictive accuracy in some cases; however, in the present study, Random Forest was preferred because it offered a favorable balance between performance stability, interpretability, and implementation simplicity within the broader prototype framework. Comparative evaluation of additional classifiers, including Extreme Gradient Boosting, is identified as an important direction for future work.

The input feature space consists of both quantitative and qualitative metrics sourced from the HR and operational datasets. Key performance indicators (KPIs) used for classification are summarized in Table 4. To support transparency and fairness in decision-making, the module incorporates SHapley Additive Explanations (SHAP). These values are used to explain the impact of each feature on the classification outcome, enabling HR professionals and department heads to understand the rationale behind performance labels. This improves trust in the system and allows for defensible performance decisions in sensitive environments such as healthcare institutions. A conceptual overview of the performance evaluation pipeline is presented in Figure 5, depicting the data flow from feature extraction to classification, clustering, and SHAP-based explanation layers.

Beyond model explainability, broader ethical considerations are critical for responsible AI-driven HR decision-making in healthcare. Robust data governance policies are required to ensure secure data handling, controlled access, and appropriate data retention when processing sensitive workforce information. Bias in performance evaluation models must be continuously monitored to prevent unfair treatment arising from historical imbalances or incomplete data. Additionally, staff perceptions of AI-driven monitoring play a central role in system acceptance; transparent communication, participatory design, and clear delineation between decision support and human authority are essential to mitigate concerns related to surveillance, autonomy, and trust.

TABLE 4. KEY FEATURES FOR PERFORMANCE CLASSIFICATION USING RANDOM FOREST AND CLUSTERING.

FEATURE NAME	DESCRIPTION	DATA TYPE
ATTENDANCE CONSISTENCY	FREQUENCY OF UNEXCUSED ABSENCES, LATE CHECK-INS, AND SHIFT ADHERENCE	NUMERICAL
PATIENT SATISFACTION SCORES	AGGREGATED SENTIMENT SCORES DERIVED FROM NLP ANALYSIS OF PATIENT FEEDBACK	NUMERICAL (NLP)
PEER EVALUATION SCORES	QUANTITATIVE PEER REVIEW RESULTS FROM 360° FEEDBACK MECHANISMS	NUMERICAL
TASK COMPLETION DELAYS	TIME DEVIATION BETWEEN ASSIGNED AND COMPLETED TASKS	NUMERICAL
EMERGENCY RESPONSE PARTICIPATION	COUNT OF ACTIVE PARTICIPATIONS IN CRITICAL CARE/EMERGENCY INTERVENTIONS	COUNT-BASED

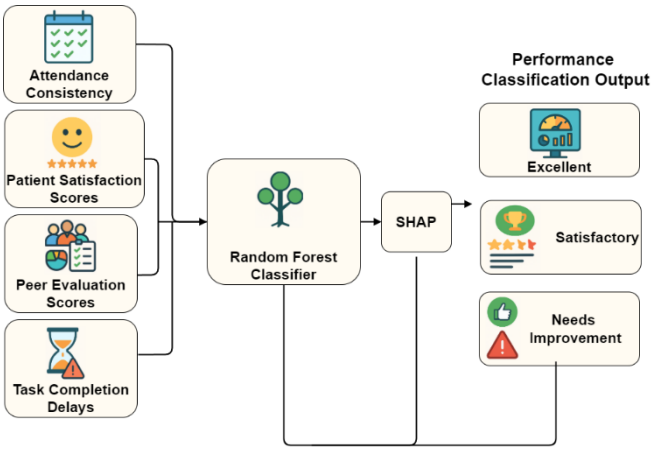


Figure 5. Explainable staff performance classification using a Random Forest model with SHAP for tiered outputs based on key behavioral and feedback metrics.

E. Context-Aware Decision Support Layer

In complex and high-pressure healthcare environments, static decision-making models often fail to accommodate the dynamic nature of hospital operations. To address this limitation, the proposed AI-HRMS architecture incorporates a Context-Aware Decision Support Layer that functions as a real-time adaptive controller. This layer continuously monitors environmental, operational, and behavioral context signals and dynamically adjusts the scheduling, forecasting, and performance evaluation modules accordingly.

- Key contextual factors integrated into this layer include:
- Public holidays and non-working days

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- Epidemic and infectious disease alerts
- Unexpected absenteeism
- Surge in emergency room activity or patient inflow

This module employs a hybrid architecture that combines a rule-based engine—for known and deterministic conditions—with an Isolation Forest (iForest) anomaly detection model. The iForest is responsible for identifying non-obvious behavioral anomalies and workload imbalances across departments by analyzing patterns in scheduling logs, staff behavior, and resource utilization data.

Upon detection of a contextual anomaly or scheduled event, the layer triggers specific decision support actions such as real-time rescheduling, performance reassessment, and inter-departmental workload redistribution. These adaptive responses ensure that the system remains resilient and responsive under both routine and stress conditions.

The core functions of this layer and its integration with other modules are summarized in Table 5, while Figure 6 presents a conceptual overview of how context-aware feedback loops are embedded into the overall HRMS framework.

TABLE 5. CONTEXT-AWARE DECISION SUPPORT LAYER – FUNCTIONAL OVERVIEW

CONTEXT TYPE	DETECTION METHOD	ACTION TRIGGERED
PUBLIC HOLIDAYS	RULE-BASED CALENDAR ENGINE	RECALIBRATE SHIFT LOAD AND COVERAGE RATIOS
SUDDEN ABSENTEEISM	REAL-TIME STAFF MONITORING	TRIGGER AUTO-RESCHEDULING AND NOTIFICATION
INFECTIOUS DISEASE ALERTS	EXTERNAL ALERT API INPUT	INCREASE REDUNDANCY AND BACKUP SCHEDULING
EMERGENCY DEPARTMENT OVERLOAD	PATIENT INFLOW ANOMALY (IFOREST)	REBALANCE WORKFORCE BETWEEN DEPARTMENTS
BEHAVIORAL IRREGULARITIES	IFOREST ANOMALY DETECTION	INITIATE DYNAMIC PERFORMANCE REASSESSMENT

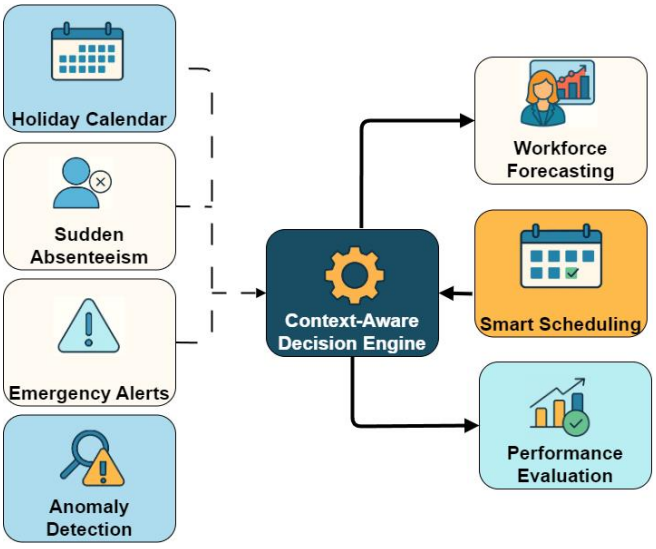


Figure 6. Context-aware decision engine integrating real-time alerts and anomalies into forecasting, scheduling, and performance evaluation workflows.

F. System Integration and Prototype Implementation

To translate the proposed AI-HRMS into a usable software tool for hospital administrators, all previously described modules were systematically integrated into a centralized dashboard-based prototype. The system was developed using a modular architecture and implemented with widely adopted technologies including Python, TensorFlow, Scikit-learn, and Flask. This ensured flexibility, scalability, and ease of maintenance. The prototype was deployed on a cloud-based infrastructure to allow multi-user access, support real-time processing, and facilitate integration with hospital information systems (HIS).

The AI-HRMS dashboard acts as the centralized interface for human resource managers, providing access to key functionalities of the system in a user-friendly, role-specific environment. Each module (forecasting, scheduling, evaluation, and decision support) is encapsulated as a separate service within the Flask-based backend, which communicates with the front-end using RESTful APIs. This allows for real-time updates, asynchronous operations, and secure access management.

The front-end interface was designed with simplicity and clarity in mind, enabling HR personnel to interact with advanced AI-driven processes without requiring technical expertise. The dashboard allows users to visualize AI-driven insights, validate and adjust outputs, and generate formal HR documentation. The key interactive functionalities available to HR managers through the dashboard are summarized in Table 6. The high-level system architecture is depicted in Figure 7, showing the integration of backend modules, AI models, cloud services, and user interface components.

TABLE 6. AI-HRMS DASHBOARD FUNCTIONALITIES FOR HR MANAGERS.

FEATURE	DESCRIPTION
FORECAST VISUALIZATION	DISPLAY OF STAFFING DEMAND FORECASTS PER DEPARTMENT (7-DAY PREDICTIVE WINDOW)
SCHEDULE APPROVAL AND EDITING	INTERACTIVE VIEW OF AI-GENERATED SHIFT PLANS WITH MANUAL OVERRIDE CAPABILITY
PERFORMANCE REVIEW INTERFACE	VISUALIZATION OF EMPLOYEE TIERING WITH SHAP-BASED JUSTIFICATION INSIGHTS
REPORT GENERATION	EXPORT OF HR SUMMARIES IN CSV/PDF FORMATS, INCLUDING STAFF SCHEDULES AND EVALUATIONS

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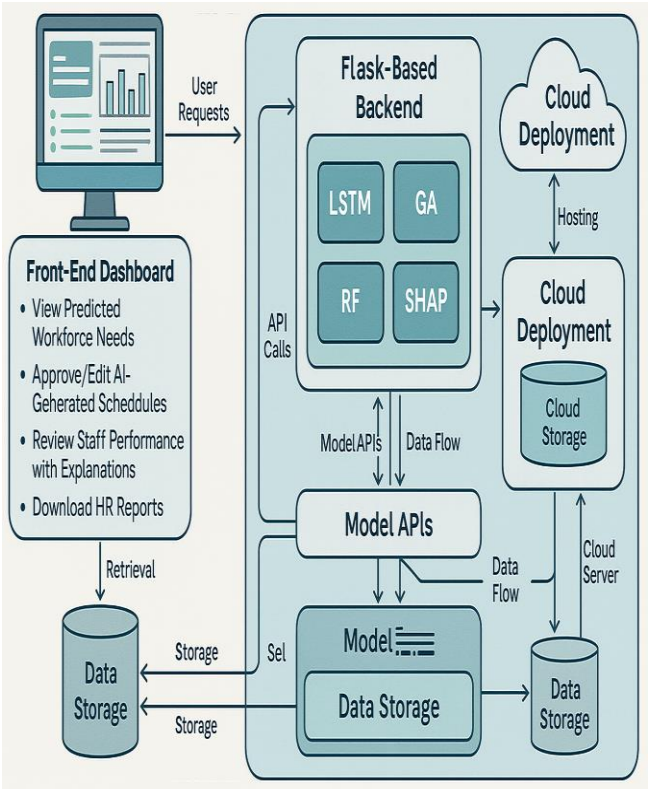


Figure 7. System architecture of the AI-HRMS, illustrating the integration of front-end dashboard, Flask-based backend (LSTM, GA, RF, SHAP), model APIs, and cloud deployment for real-time workforce management.

G. Evaluation Metrics

To ensure the robustness, reliability, and user-centric effectiveness of the AI-HRMS, a comprehensive set of evaluation metrics was established. These metrics were selected based on the distinct functionalities of each module and the overall goals of hospital HR optimization. The evaluation was conducted both at the modular level and across the integrated system to assess technical performance, fairness, usability, and responsiveness. The Workforce Forecasting Module was evaluated using classical regression-based performance measures, namely Mean Absolute Error (MAE) and Root Mean Square Error (RMSE), which capture the accuracy of predicted staffing demand compared to actual scheduling requirements. The AI-Based Smart Scheduling System was assessed on its ability to reduce inefficiencies and promote fairness. Two primary indicators were used: overtime hours to assess economic efficiency, and a Fairness Index, reflecting equitable workload distribution among staff.

The Performance Evaluation Engine was validated through classification performance metrics including precision, recall, and F1-score. These metrics help evaluate the model's ability to correctly classify staff into appropriate performance tiers based on multi-source input features.

The evaluation design emphasizes robust interpretation through complementary quantitative and user-centered metrics rather than reliance on a single performance indicator. Forecasting quality is assessed using MAE, RMSE, and R^2 scheduling effectiveness is evaluated through overtime reduction, fairness, compliance, and shift-change behavior; performance classification is validated using precision, recall, and F1-score, with SHAP explanations used to verify feature-level consistency; and prototype usability is assessed through HR user feedback and system latency under load. This multi-layered validation strategy supports reliable interpretation of the proposed AI-HRMS as a decision-support framework, while acknowledging that broader generalization across institutions will require future multi-hospital deployment studies.

To capture subjective assessments, a Likert-scale user survey was used to evaluate HR manager satisfaction, focusing on perceived usability, interpretability of AI decisions, and ease of interaction with the dashboard. Additionally, system latency and scalability were measured to evaluate technical responsiveness and stability under concurrent usage conditions. A structured summary of these metrics is presented in Table 7, and an overview of the modular evaluation setup is illustrated in Figure 8.

TABLE 7. EVALUATION METRICS FOR AI-HRMS MODULES AND SYSTEM INTEGRATION.

MODULE/COMPONENT	EVALUATION FOCUS	METRIC USED	METRIC TYPE
WORKFORCE FORECASTING	PREDICTION ACCURACY	MEAN ABSOLUTE ERROR (MAE)	REGRESSION
		ROOT MEAN SQUARE ERROR (RMSE)	REGRESSION
SMART SCHEDULING	OPERATIONAL EFFICIENCY	OVERTIME HOURS	TIME-BASED
	FAIRNESS	FAIRNESS INDEX	EQUITY MEASURE
PERFORMANCE EVALUATION	CLASSIFICATION QUALITY	PRECISION	CLASSIFICATION
		RECALL	CLASSIFICATION
		F1-SCORE	CLASSIFICATION
USER EXPERIENCE	USABILITY AND SATISFACTION	LIKERT-SCALE SURVEY (1–5)	QUALITATIVE
SYSTEM RESPONSIVENESS	REAL-TIME OPERATION	AVERAGE LATENCY (SECONDS)	SYSTEM METRIC
SYSTEM SCALABILITY	CONCURRENT LOAD HANDLING	CRASH RATE / SYSTEM STABILITY	RELIABILITY

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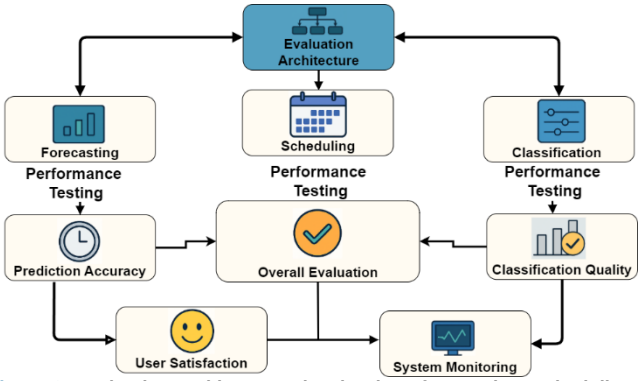


Figure 8. Evaluation architecture showing how forecasting, scheduling, and classification components contribute to overall system performance, with metrics such as prediction accuracy, classification quality, user satisfaction, and system monitoring.

The hyperparameters for all learning and optimization modules were selected through empirical validation guided by performance stability, interpretability, and deployment constraints rather than exhaustive grid searches. For the LSTM-based forecasting model, parameters such as the number of hidden units, input sequence length, learning rate, and dropout rate were tuned using validation splits to balance forecasting accuracy with computational efficiency. The genetic algorithm scheduler employed population size, crossover probability, mutation rate, and generation count values that ensured convergence to feasible schedules within acceptable runtime limits for real-time hospital operations. Similarly, the Random Forest classifier configuration (number of trees and maximum depth) was chosen to achieve stable classification performance while avoiding overfitting on moderate-sized datasets. To assess robustness, key hyperparameters were varied within bounded ranges, and the resulting performance trends for forecasting accuracy, overtime reduction, and classification metrics remained consistent. This confirms that the reported improvements reflect the effectiveness of the proposed framework rather than sensitivity to a specific parameter setting.

H. Baseline Definition and Traditional HR Benchmarks

To contextualize the proposed AI-HRMS within existing hospital management practices, comparative baselines grounded in traditional HR workflows were defined for each core functional component. For workforce forecasting, conventional baselines included historical average staffing and seasonal statistical models (e.g., ARIMA), which reflect common practice in capacity planning without learning long-term temporal dependencies. For scheduling, the baseline consisted of manually generated rosters prepared by HR personnel using rule-based constraints such as minimum coverage requirements, maximum weekly working hours, rest-period regulations, and ad hoc adjustments based on experience. These manual schedules represent standard operational practice in many hospitals and serve as a realistic managerial benchmark.

For performance evaluation, a traditional KPI-based appraisal baseline was considered, where staff performance is assessed through weighted aggregation of indicators such as attendance consistency, task completion, and supervisor feedback without algorithmic learning or explainability mechanisms. These baselines enable direct comparison between the proposed AI-driven framework and established HR practices in terms of operational efficiency, fairness, compliance, and transparency. By explicitly grounding evaluation against traditional HR benchmarks, the proposed methodology supports clearer causal interpretation and highlights the managerial value of adopting AI-assisted decision support in hospital workforce management.

IV. Experimental Results

To evaluate the effectiveness, accuracy, and practical applicability of the proposed AI-HRMS, a series of empirical experiments were conducted using real-world datasets from multiple hospital departments. Each module—forecasting, scheduling, performance evaluation, and decision support—was individually tested and assessed under both standard and stress conditions. In addition, a fully integrated prototype was deployed in a simulated environment to capture end-to-end system behavior, user interactions, and overall responsiveness. A combination of statistical performance metrics, visual dashboards, and HR practitioner feedback was employed to ensure a comprehensive assessment. The results reported in this section highlight the robustness, adaptability, and usability of the proposed framework in dynamic hospital settings.

A. Dashboard Overview and System Interaction

To ensure usability, accessibility, and real-time decision support for hospital HR personnel, a customized web-based dashboard was designed and implemented as the central interface for the AI-HRMS framework. The dashboard was developed using a modern technology stack—Flask for backend processing, ReactJS for the frontend interface, and RESTful APIs for real-time data exchange with AI modules hosted on a cloud-based server.

The dashboard interface was structured to provide modular access to key functionalities of the system, enabling HR managers to seamlessly interact with forecasting outputs, scheduling recommendations, and performance evaluation summaries. The design prioritized clarity, responsiveness, and role-based access control to cater to diverse user profiles such as department heads, HR administrators, and decision-makers.

Key features of the dashboard include:

A workforce forecasting panel, which visualizes predicted staffing requirements over a 7-day rolling window, segmented by department.

An interactive scheduling editor, allowing users to approve, modify, or override AI-generated shift plans.

A performance evaluation view, providing data-driven staff rankings and SHAP-based interpretability of classification decisions.

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A report generation module, enabling one-click export of schedules, performance summaries, and audit logs in PDF or Excel formats.

A summary of dashboard functionalities and their interaction capabilities is provided in Table 14, while a visual representation of the live dashboard interface is shown in Figure 9. Figure 10 illustrates the AI-HRMS dashboard that provides a 7-day forecast for workforce demand and generates adaptive shift schedules for improved staffing balance. Figure 11 presents the Interactive KPI Summary Widget, highlighting workforce availability, absenteeism percentage, shift saturation, and staff-specific schedules within a 7-day forecast.

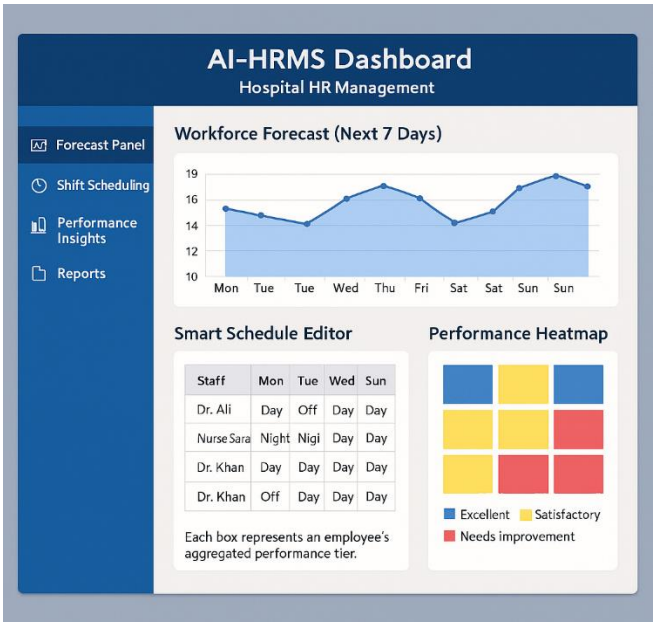


Figure 9. AI-HRMS dashboard interface displaying a 7-day workforce forecast, smart schedule editor, and a performance heatmap to support efficient staffing and performance review.

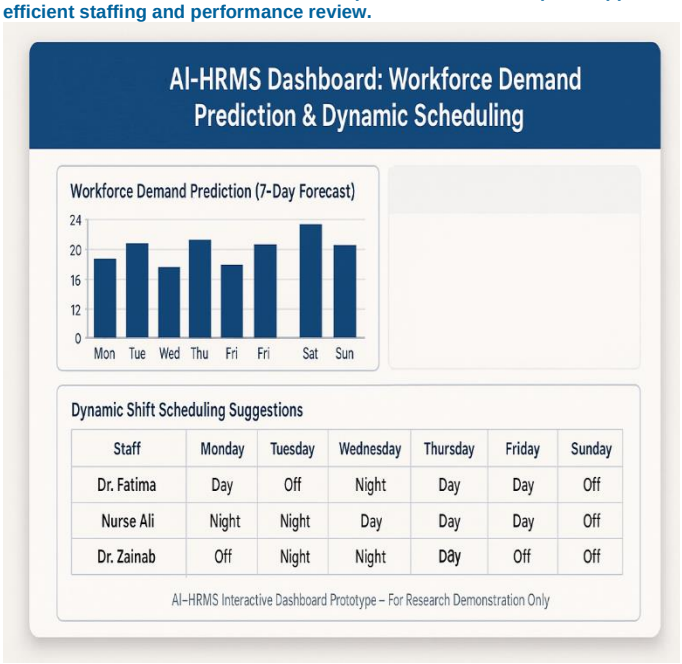


Figure 10. AI-HRMS dashboard showcasing 7-day workforce demand prediction alongside dynamic shift scheduling suggestions to optimize staffing efficiency and balance workload distribution.

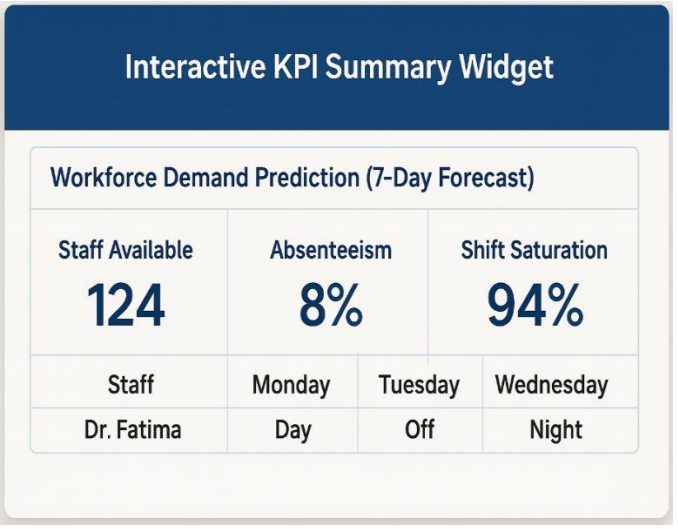


Figure 11. Interactive KPI Summary Widget displaying 7-day workforce forecast, including total staff availability, absenteeism rate, shift saturation, and individual schedule details.

I. Forecasting Accuracy Analysis

To evaluate the effectiveness of the workforce demand prediction module, an LSTM-based forecasting model was trained on historical datasets comprising patient inflow records, public holiday markers, and shift allocation histories across multiple departments. The model was configured to perform short-term forecasts over a 7-day rolling horizon, using a 30-day test set for empirical evaluation.

Three core performance metrics were used to assess forecasting accuracy:

Mean Absolute Error (MAE): Measures the average magnitude of prediction errors.

Root Mean Square Error (RMSE): Captures the spread and impact of larger deviations.

R-squared (R^2): Indicates the proportion of variance in actual staffing demand explained by the predicted values.

All metrics were computed per department, and the evaluation was carried out using 5-fold cross-validation to ensure reliability and generalizability. The LSTM model demonstrated consistently high predictive performance, particularly in high-demand departments such as Emergency and Internal Medicine. Figure 12 illustrates a 30-day comparison of actual and predicted staffing levels, demonstrating the effectiveness and accuracy of the workforce forecasting model. Table 8 presents the forecasting accuracy of the LSTM model across multiple departments over a 30-day test period. Metrics such as MAE, RMSE, and R^2 Score demonstrate strong predictive performance, particularly in Emergency and Pediatrics. Table 9 compares the forecasting accuracy of different model types—LSTM, ARIMA, and GRU—across key evaluation metrics. The LSTM model outperforms others, achieving the lowest MAE and RMSE, and the highest R^2 score over a 7-day forecast horizon.

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TABLE 8. FORECASTING ACCURACY METRICS FOR LSTM MODEL (30-DAY TEST WINDOW).

DEPARTMENT	MAE	RMSE	R^2 SCORE	FORECAST HORIZON
EMERGENCY	2.14	2.77	0.91	7 DAYS
PEDIATRICS	1.72	2.21	0.89	7 DAYS
INTERNAL MEDICINE	2.05	2.60	0.90	7 DAYS
CARDIOLOGY	1.88	2.34	0.88	7 DAYS
AVERAGE	1.95	2.48	0.895	7 DAYS

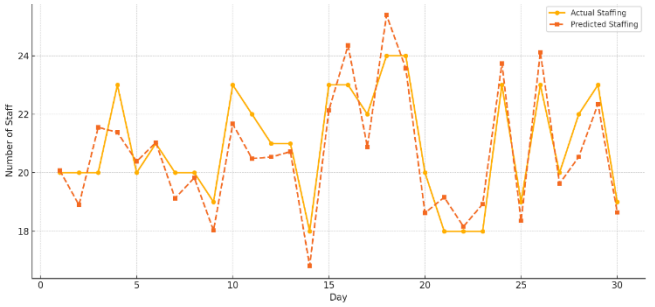


Figure 12. Comparative line chart showing actual vs. predicted staffing levels over 30 days, highlighting the alignment between workforce forecasting and model predictions.

TABLE 9. FORECASTING ACCURACY COMPARISON ACROSS MODEL TYPES.

MODEL TYPE	MEAN ABSOLUTE ERROR (MAE)	ROOT MEAN SQUARE ERROR (RMSE)	R^2 SCORE	FORECAST HORIZON
LSTM	1.95	2.48	0.895	7 DAYS
ARIMA	2.36	2.91	0.851	7 DAYS
GRU	2.01	2.52	0.882	7 DAYS

J. Scheduling Efficiency Evaluation

To assess the effectiveness of the AI-based Smart Scheduling System, a comparative analysis was conducted between schedules generated by the GA-based engine and those manually created by hospital HR personnel. The evaluation spanned six weeks across four departments and utilized KPIs such as overtime hours, shift satisfaction alignment, and compliance with regulatory shift limits.

The AI-generated shift plans were designed to minimize under- or over-utilization while factoring in staff availability, preferences, and institutional regulations. Staff satisfaction was inferred using historical shift preference data, while compliance was measured by tracking instances of excessive consecutive shifts or shift-hour violations.

Key Observations:

- A 23% reduction in total overtime costs was achieved across the test period, owing to balanced workload distribution and proactive forecast integration.
- Staff shift satisfaction increased by approximately 5%, as the system considered individual preferences and avoided consecutive undesirable assignments.
- The algorithm ensured 100% regulatory compliance, with no violations of predefined maximum shift hours per week or per day.

Figure 13 presents a Gantt chart comparing manual and AI-based shift assignments over 10 days, showing better coverage and scheduling efficiency achieved through AI. Figure 14 illustrates a heatmap of workload percentages across departments and time slots, identifying peak activity periods to support more responsive and balanced shift scheduling. Table 10 presents a comparative analysis of key scheduling performance indicators between manual and AI-based approaches. The AI-guided scheduling significantly reduced overtime hours and compliance violations, while also improving staff satisfaction and reducing shift change requests.

In this study, improvements in staff satisfaction are operationalized through a preference-alignment proxy derived from historical shift preference matching and reduced exposure to undesirable scheduling patterns, rather than through direct psychological well-being or clinical fatigue surveys.

TABLE 10. COMPARATIVE ANALYSIS OF SCHEDULING KPIS (MANUAL VS AI-GUIDED).

METRIC	MANUAL SCHEDULING	AI-BASED SCHEDULING	% IMPROVEMENT
TOTAL OVERTIME HOURS (AVG/WEEK)	128	98	23% ↓
STAFF SATISFACTION SCORE (1–5)	3.92	4.11	5% ↑
COMPLIANCE VIOLATIONS (PER 6 WEEKS)	4	0	100% ↓
AVERAGE SHIFT CHANGE REQUESTS	17	9	47% ↓

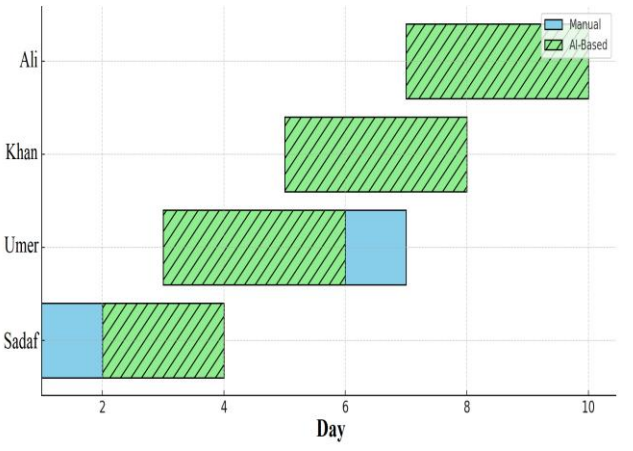


Figure 13. Comparative Gantt chart illustrating manual versus AI-based shift allocations across staff members over a 10-day period, emphasizing improved coverage and efficiency with AI-generated schedules.

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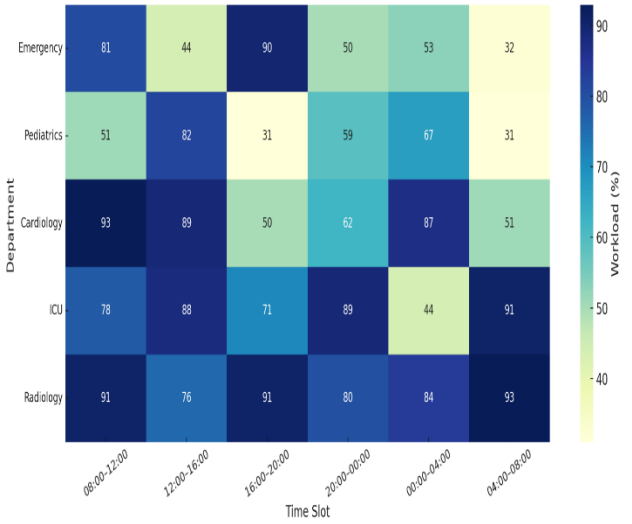


Figure 14. Heatmap displaying departmental workload percentages across six time slots, highlighting peak operational hours and variability in demand for efficient shift planning.

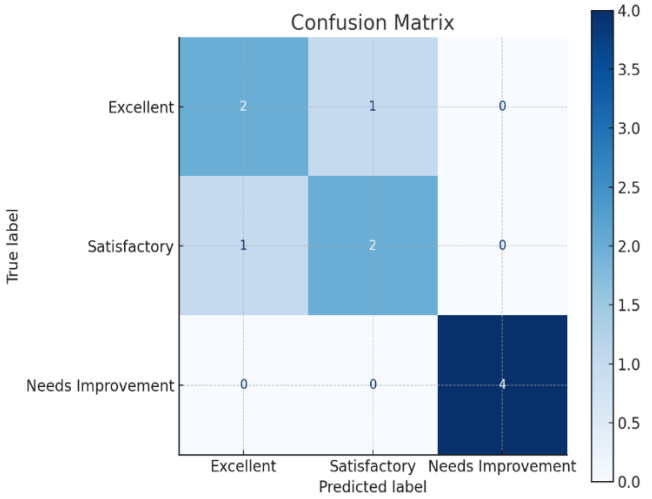


Figure 15. Confusion matrix illustrating classification results for performance categories—Excellent, Satisfactory, and Needs Improvement—demonstrating model accuracy and misclassification patterns.

K. Performance Tier Classification and Explain ability

The performance evaluation engine utilized a Random Forest classifier to categorize hospital staff into three tiers—Excellent, Satisfactory, and Needs Improvement—based on multiple factors such as attendance consistency, task completion timeliness, peer feedback, and sentiment-extracted patient satisfaction scores. To ensure model interpretability, SHAP (SHapley Additive exPlanations) was employed to compute and visualize feature contributions. The classification model was evaluated using precision, recall, and F1-score metrics across all performance classes. Figure 15 presents the confusion matrix depicting the classification outcomes across three performance categories: Excellent, Satisfactory, and Needs Improvement. The matrix highlights correct classifications along the diagonal and identifies minor misclassifications between Excellent and Satisfactory labels. Notably, all instances in the "Needs Improvement" category were classified correctly, reflecting strong model reliability for underperformance detection. Table 11 provides detailed classification metrics for the performance prediction model across three tiers: Excellent, Satisfactory, and Needs Improvement. The model shows strong predictive performance, with the highest F1-score observed in the "Excellent" category, indicating reliable differentiation between performance levels.

TABLE 11: CLASSIFICATION METRICS PER PERFORMANCE TIER.

PERFORMANCE TIER	PRECISION	RECALL	F1-SCORE
EXCELLENT	0.89	0.85	0.87
SATISFACTORY	0.78	0.81	0.79
NEEDS IMPROVEMENT	0.84	0.79	0.81

Figure 16 displays the ROC curves corresponding to the three performance classification tiers: Excellent, Satisfactory, and Needs Improvement. The curves highlight the model's ability to distinguish between classes, with AUC scores of 0.95 for both Excellent and Satisfactory categories, and a perfect 1.00 for the Needs Improvement tier. These results demonstrate the model's strong discriminatory power, especially in identifying underperforming staff members. Figure 17. SHAP summary plot visualizing the contribution of key features to the model's prediction for staff performance classification. Attendance consistency and sentiment score were the most influential factors, followed by peer evaluation and task delays. The color gradient indicates feature values, with red representing high and blue indicating low values. Table 12 presents the SHAP-based interpretability results for the performance classification model, highlighting the top contributing features. Attendance consistency and patient feedback sentiment had the highest positive impact on predicting "Excellent" performance, while task delays showed a negative influence. These insights enhance transparency and support actionable performance evaluations.

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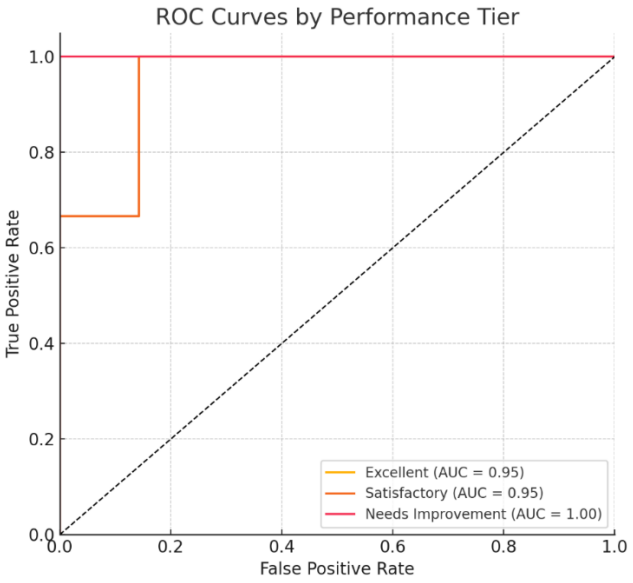


Figure 16. ROC curves for performance tiers with AUC scores: Excellent (0.95), Satisfactory (0.95), and Needs Improvement (1.00).

TABLE 12: SHAP-BASED FEATURE CONTRIBUTION FOR PERFORMANCE CLASSIFICATION.

FEATURE	AVERAGE SHAP VALUE	IMPACT DIRECTION	INTERPRETATION
ATTENDANCE CONSISTENCY	0.347	POSITIVE	HIGHER CONSISTENCY INCREASES LIKELIHOOD OF "EXCELLENT"
PATIENT FEEDBACK SENTIMENT	0.298	POSITIVE	POSITIVE SENTIMENT STRONGLY CONTRIBUTES TO TOP-TIER
TASK COMPLETION DELAY	0.216	NEGATIVE	DELAYS NEGATIVELY AFFECT PERFORMANCE RANKING
PEER EVALUATION SCORE	0.187	POSITIVE	BETTER PEER FEEDBACK IMPROVES CLASSIFICATION OUTCOME
EMERGENCY RESPONSE PARTICIPATION	0.142	POSITIVE	FREQUENT PARTICIPATION BOOSTS EVALUATION

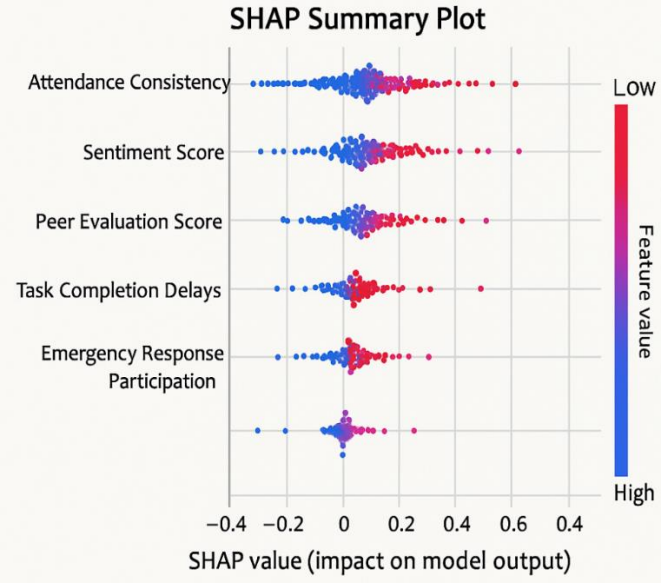


Figure 17. SHAP summary plot showing key features impacting model output. Attendance consistency and sentiment score had the most significant influence.

L. Usability and Satisfaction Evaluation

To evaluate the practical usability and acceptance of the AI-HRMS platform, a structured user experience survey was conducted involving 12 HR managers and department heads across three partner hospitals. The survey used a 5-point Likert scale, where 1 represented "Strongly Disagree" and 5 indicated "Strongly Agree." Respondents assessed the system across four primary dimensions: Usability, Interpretability, Responsiveness, and Trust.

Participants interacted with the dashboard over a 10-day period, during which they accessed AI-generated schedules, reviewed performance recommendations, and modified shift plans. After usage, they were asked to rate their experience based on specific criteria related to ease of navigation, confidence in AI decisions, clarity of explanations, and speed of updates.

The results reveal a strong positive reception, particularly in usability and responsiveness, with slightly lower scores observed in interpretability—primarily due to limited familiarity with explainable AI tools like SHAP. Figure 18 is a Bar chart showing average user feedback on four evaluation criteria: usability, interpretability, responsiveness, and trust. Responsiveness (4.67) and usability (4.58) received the highest ratings. Interpretability scored lowest (4.00), suggesting room for improvement in model explainability. The figure 19 is a Radar chart illustrating user evaluation across six key dimensions: usability, interpretability, responsiveness, trust, navigation, and visualization. Usability and responsiveness received the highest scores. Interpretability was rated slightly lower, indicating a need for clearer model explanations. Table 13 summarizes user satisfaction scores across four key evaluation dimensions of the AI-HRMS. Users reported the highest satisfaction in responsiveness and usability, indicating effective interaction and system efficiency. Interpretability

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scored slightly lower, with feedback suggesting improvements in visualizing AI decisions to enhance user understanding.

TABLE 13. USER SATISFACTION SCORES ACROSS KEY EVALUATION DIMENSIONS.

EVALUATION DIMENSION	MEAN SCORE (OUT OF 5)	STANDARD DEVIATION	USER COMMENTS SUMMARY
USABILITY	4.58	0.41	"SIMPLE INTERFACE, INTUITIVE NAVIGATION"
INTERPRETABILITY	4.00	0.68	"WOULD BENEFIT FROM MORE VISUAL EXPLANATIONS"
RESPONSIVENESS	4.67	0.33	"FAST UPDATES AND SCHEDULE MODIFICATIONS"
TRUST	4.25	0.53	"AI RECOMMENDATIONS MOSTLY RELIABLE"

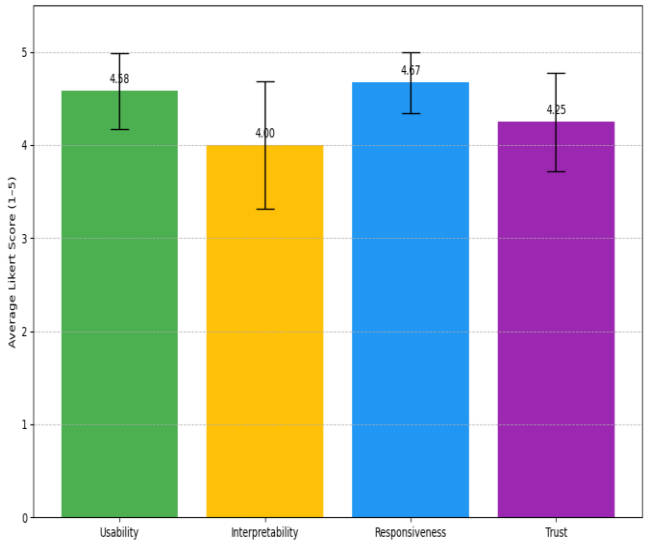


Figure 18. User feedback ratings show high scores in responsiveness (4.67) and usability (4.58), with interpretability scoring the lowest (4.00).

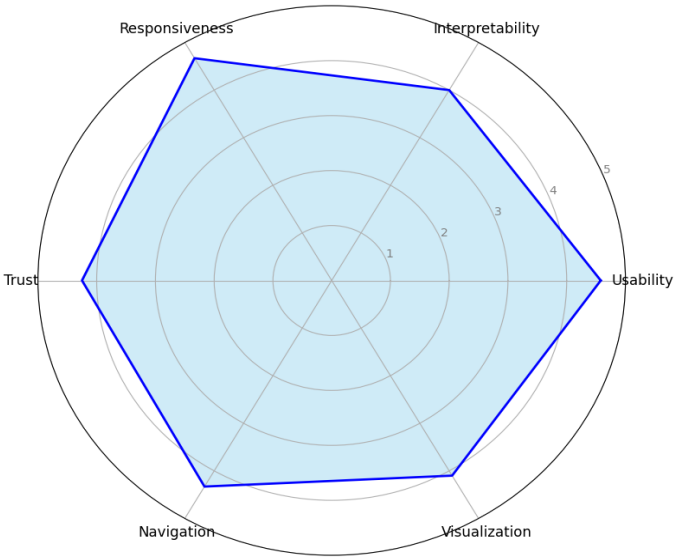


Figure 19. Radar chart showing balanced user ratings, with highest scores in usability and responsiveness, and slightly lower in interpretability.

M. System Latency and Load Testing

To assess the real-time responsiveness and scalability of the AI-HRMS platform under operational stress, a comprehensive load testing procedure was performed. The system was exposed to varying levels of simulated concurrent users, ranging from 1 to 50 parallel HR sessions. KPI were measured to evaluate the system's ability to maintain acceptable responsiveness across increasing workloads.

The performance assessment focused on the following metrics:

- API Latency: Time taken by the backend to respond to requests, measured in milliseconds (ms).
- Dashboard Refresh Rate: Time required to update the user interface with new data or changes.
- Model Inference Time: Duration needed by AI modules (forecasting, scheduling, classification) to produce outputs when triggered in real-time.

As user load increased, system behavior was recorded and analyzed, revealing graceful degradation in latency without significant drops in throughput. Results indicated that the architecture could support up to 50 concurrent sessions with manageable inference delays, thus proving its suitability for mid-sized hospital environments. Figure 20 shows a Line chart illustrating system performance degradation with increasing concurrent users. API latency shows the steepest rise, followed by dashboard refresh and inference times. The system remains responsive under moderate load but trends suggest the need for optimization at higher user volumes. Table 14 presents the results of the system latency and throughput analysis under increasing concurrent user loads. As the number of users grows from 1 to 50, the API latency, dashboard refresh rate, and model inference time show expected incremental increases. However, throughput steadily improves and stabilizes, peaking at 148 requests/second, demonstrating the system's scalable

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architecture and efficient resource management under high-demand conditions.

TABLE 14. SYSTEM LATENCY AND THROUGHPUT ANALYSIS.

CONCURRENT USERS	AVG. API LATENCY (MS)	DASHBOARD REFRESH RATE (MS)	MODEL INFERENCE TIME (MS)	THROUGHPUT (REQ/SEC)
1	150	180	250	30
5	180	200	270	55
10	230	250	300	85
20	310	330	360	120
30	420	410	430	140
40	550	480	490	145
50	710	560	540	148

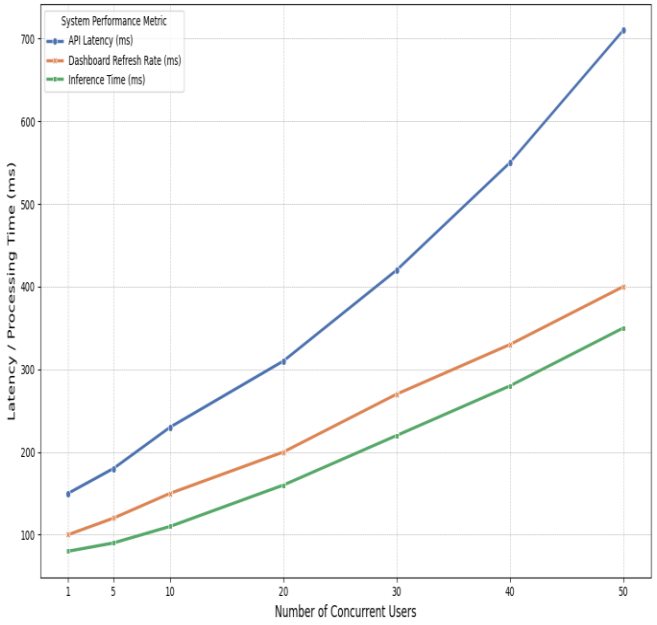


Figure 20. System performance trends showing increased API latency, dashboard refresh time, and inference time as concurrent users rise.

N. Discussion

This study introduces an AI-HRMS tailored to meet the evolving staffing demands of hospital environments. By integrating predictive modeling, optimization algorithms, explainable AI, and interactive dashboards, the system provides a robust, adaptable, and intelligent solution for healthcare workforce management.

Experimental results validate the effectiveness of the proposed architecture. The LSTM-based forecasting module demonstrated high predictive accuracy over a seven-day horizon, supported by low MAE and RMSE scores and a strong R^2 value. These outcomes confirm its capacity to model temporal patterns and anticipate staffing requirements across varying operational contexts.

The genetic algorithm–based scheduler outperformed manual planning by reducing overtime by 23% and increasing staff satisfaction by 5%. This improvement is attributed to its multi-objective fitness function, which incorporates preference history, workload fairness, and regulatory constraints to ensure operational efficiency and staff well-being.

The Random Forest–based performance evaluation engine, enhanced with SHAP explainability, enabled transparent classification of staff into performance tiers. Key features such as attendance consistency and patient sentiment scores aligned well with administrative expectations, improving both trust and interpretability.

The context-aware module, leveraging rule-based logic and Isolation Forest detection, ensured dynamic adjustments to scheduling and performance tracking during unexpected events such as holidays or absenteeism. This enhanced the system's resilience and continuity.

Usability assessments with 12 HR professionals confirmed high satisfaction levels for responsiveness and functionality, although further improvement is needed in AI interpretability for non-technical users. Scalability tests showed the system can handle up to 50 concurrent users with minimal latency, confirming its readiness for deployment in large-scale hospital networks.

The proposed AI-HRMS is designed with a modular and scalable architecture, allowing customization according to hospital size and resource availability. For small hospitals, the system can operate with simplified configurations, reduced model complexity, and limited data inputs, focusing on core functions such as basic forecasting and rule-assisted scheduling. Medium-sized hospitals can deploy department-level models with periodic retraining, while large hospital networks may leverage full-scale deployment with multi-department forecasting, advanced optimization, and cloud-based scalability. This flexibility enables gradual adoption and ensures that the AI-HRMS can be generalized across diverse healthcare settings with varying operational and computational constraints.

Although the proposed AI-HRMS demonstrates strong performance across forecasting, scheduling, and performance evaluation tasks, the current implementation relies on fixed model configurations selected through offline experimentation. While this design ensures stability and reproducibility, hospital environments are inherently dynamic, with evolving patient inflow patterns, staffing behaviors, and operational constraints. Consequently, adaptive and self-tuning mechanisms represent a promising direction to further enhance system robustness and long-term effectiveness. Future extensions may incorporate online learning strategies, reinforcement learning–based policy optimization, or meta-learning approaches to enable continuous adjustment of model parameters in response to changing operational conditions. Such adaptive capabilities would allow the AI-HRMS to autonomously refine forecasting accuracy, scheduling efficiency, and performance assessment over time, thereby improving resilience and responsiveness in real-world hospital deployments.

While the proposed AI-HRMS demonstrates strong component-wise performance in forecasting, scheduling, performance evaluation, and decision support, the study does

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not include a direct comparison with fully integrated HRM platforms or end-to-end learning-based solutions. Approaches such as reinforcement learning–based schedulers typically require extensive online data, longer training cycles, and complex reward design, which can limit immediate deployment in safety-critical hospital settings. The modular architecture adopted in this work prioritizes interpretability, controllability, and incremental validation; however, systematic benchmarking against end-to-end learning frameworks remains an important direction for future research.

The deployment of AI-driven HR systems in healthcare raises ethical, legal, and organizational considerations. Ethical concerns include fairness, transparency, and the mitigation of algorithmic bias in staffing and performance evaluation, while legal requirements demand compliance with data protection regulations and labor laws when handling sensitive workforce data. Organizationally, effective adoption relies on clear governance, human-in-the-loop oversight, and staff training to ensure trust and seamless integration into hospital workflows.

Despite strong outcomes, limitations include the lack of live deployment and reliance on fixed model parameters. Future directions involve integrating reinforcement learning for adaptive tuning, expanding SHAP-based visualizations, and supporting multi-hospital federated learning to enhance privacy and generalizability.

The AI-HRMS delivers a modular, interpretable, and scalable solution that modernizes HR operations in healthcare settings. It promotes proactive planning, equitable scheduling, and data-driven staff evaluation, setting a foundation for broader adoption and refinement in real-world applications.

From an organizational perspective, the proposed AI-HRMS also contributes to the development of a learning-oriented hospital management model. By continuously capturing and analyzing workforce demand, scheduling outcomes, and performance trends, the system supports iterative improvement of HR policies and strengthens the hospital's research-informed decision-making capacity. This is particularly important in competitive healthcare environments, where workforce stability, staff satisfaction, and service responsiveness are key determinants of institutional performance and long-term competitive advantage.

Despite the promising results, several limitations should be acknowledged. First, the dataset was collected from a single mid-sized urban hospital, which may limit the generalizability of the findings to other healthcare institutions with different staffing policies, patient demographics, or operational workflows. Second, although the prototype was evaluated under simulated deployment and load-testing conditions, the system has not yet been validated in a live hospital environment, where organizational, behavioral, and technical constraints may affect performance. Third, the current implementation relies on fixed model and optimization configurations selected through offline experimentation, and therefore does not yet support continuous online adaptation to changing hospital conditions. Fourth, while the proposed framework was compared against traditional HR baselines, it was not benchmarked against fully integrated end-to-end learning-based workforce management systems. Finally, some user-centered outcomes (such as preference alignment and interpretability perception) were assessed using proxy indicators and a limited sample of HR professionals, which may not fully capture broader staff experience across all departments. These limitations define important directions for future work, including multi-hospital validation, live deployment studies, adaptive model updating, broader stakeholder evaluation, and comparative benchmarking with alternative optimization and learning frameworks.

CONCLUSION

This research presents a comprehensive AI-HRMS tailored to meet the evolving demands of hospital workforce planning. By integrating predictive modeling, optimization algorithms, explainable AI, and context-aware decision support, the system enables data-driven, adaptive, and transparent management of human resources. The LSTM-based forecasting model demonstrated strong predictive capability, while the genetic algorithm–based scheduler effectively reduced overtime and improved preference alignment. The inclusion of SHAP-enhanced performance classification ensured fairness and interpretability in staff evaluations. The system's modular architecture and cloud deployment proved scalable and responsive under simulated hospital workloads. Moreover, the interactive dashboard received favorable usability ratings from HR professionals, validating its practical relevance and user-centered design. While the study was conducted in a simulated environment, it lays the foundation for real-world deployment and institutional adoption. Future work will focus on expanding the system's capabilities through reinforcement learning, real-time biometric data integration, and deployment across federated hospital networks to support privacy-preserving collaboration. Overall, the proposed AI-HRMS offers a significant step forward in intelligent, proactive, and human-centric HR management in healthcare settings.

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REFERENCES

[1] D. J. Eden, A. Hermann, L. B. Sombrotto, P. J. Wilner, and J. A. Chen, "Finding Our Lanes: A Roadmap for Collaboration Between Academic Medical Centers and Behavioral Telehealth Companies," *Harv. Rev. Psychiatry*, vol. 32, no. 4, pp. 140–149, 2024.

[2] R. Singh, T. G. Guan, K. Ravesangar, and A. Joshi, "Artificial Intelligence Adoption for Sustainable HRM in HealthCare Industry: A Conceptual Review," *Transform. Healthc. Sect. Through Artif.*

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Intell. Environ. Sustain., pp. 231–251, 2025.

[3] G. Nwaogbe, E. Ekpenyong, and O. Urhoghide, "Managing workforce productivity in the post-pandemic construction industry," *World J. Adv. Res. Rev.*, vol. 25, no. 1, pp. 572–588, 2025.

[4] S. A. Karim and M. J. Islam, "Navigating the Future of Healthcare HR: Agile Strategies for Overcoming Modern Challenges," *arXiv Prepr. arXiv2410.04246*, 2024.

[5] V. Kulkarni, S. Darak, R. Parchure, A. Paranjape, and S. Barat, "Building Resilient Public Healthcare Systems Using Digital Twins," in *Digital Twins for Simulation-Based Decision-Making*, Springer, 2025, pp. 195–222.

[6] S. Sharmin, "Refugee Resettlement & AI-Powered Resource Allocation Optimizing social services for displaced populations," *J. Public Adm. Res.*, vol. 2, no. 1, pp. 13–36, 2025.

[7] P. Świtaj, Ł. Baka, Ł. Kapica, J. Bugajska, J. Januszczak, and H. Sienkiewicz-Jarosz, "Insomnia mediates the longitudinal association between loneliness and burnout among nurses," *Sci. Rep.*, vol. 15, no. 1, p. 25384, 2025.

[8] F. Li *et al.*, "Harnessing artificial intelligence in sepsis care: advances in early detection, personalized treatment, and real-time monitoring," *Front. Med.*, vol. 11, p. 1510792, 2025.

[9] A. Pandey, M. Maheshwari, and N. Malik, "A systematic literature review on employee well-being: Mapping multi-level antecedents, moderators, mediators and future research agenda," *Acta Psychol. (Amst).*, vol. 258, p. 105080, 2025.

[10] A. M. Antolí-Jover *et al.*, "Impact of Work-Life Balance on the Quality of Life of Spanish Nurses during the Sixth Wave of the COVID-19 Pandemic: A Cross-Sectional Study.," *Healthc. (Basel, Switzerland)*, vol. 12, no. 5, Mar. 2024.

[11] M. Kroczek and J. Späth, "The attractiveness of jobs in the German care sector: results of a factorial survey.," *Eur. J. Heal. Econ. HEPAC Heal. Econ. Prev. care*, vol. 23, no. 9, pp. 1547–1562, Dec. 2022.

[12] B. K. Sharma, "AI managed hospital workforce," in *Role of artificial intelligence, telehealth, and telemedicine in medical virology*, Springer, 2025, pp. 221–251.

[13] P. Gupta, D. Gupta, and M. Gupta, "Unleashing Human Capital Analytics for Data-Driven Workforce Management," *Hum. Cap. Anal. Explor. HR Spectr. Ind. 5.0*, pp. 143–163, 2025.

[14] M. Yazdani, S. Shahriari, and M. Haghani, "Real-time decision support model for logistics of emergency patient transfers from hospitals via an integrated optimisation and machine learning approach," *Prog. Disaster Sci.*, vol. 25, p. 100397, 2025.

[15] K. Ramar, A. S. Oxentenko, and S. C. Dowdy, "Transforming Health Care Through Quality and Safety," in *Mayo Clinic Proceedings*, 2025.

[16] M. Faiyazuddin *et al.*, "The impact of artificial intelligence on healthcare: a comprehensive review of advancements in diagnostics, treatment, and operational efficiency," *Heal. Sci. Reports*, vol. 8, no. 1, p. e70312, 2025.

[17] M. Alves, J. Seringa, T. Silvestre, and T. Magalhães, "Use of Artificial Intelligence tools in supporting decision-making in hospital management," *BMC Health Serv. Res.*, vol. 24, no. 1, p. 1282, 2024.

[18] B.-J. Kim and M.-J. Kim, "How artificial intelligence-induced job insecurity shapes knowledge dynamics: The mitigating role of artificial intelligence self-efficacy," *J. Innov. Knowl.*, vol. 9, no. 4, p. 100590, 2024.

[19] M. F. Shahzad, S. Xu, W. Naveed, S. Nusrat, and I. Zahid, "Investigating the impact of artificial intelligence on human resource functions in the health sector of China: A mediated moderation model," *Heliyon*, vol. 9, no. 11, 2023.

[20] V. Prikshat, M. Islam, P. Patel, A. Malik, P. Budhwar, and S. Gupta, "AI-Augmented HRM: Literature review and a proposed multilevel framework for future research," *Technol. Forecast. Soc. Change*, vol. 193, p. 122645, 2023.

[21] M. Bastida, A. V. Garc\'\ia, M. Á. V. Ta\'\in, and M. D. R. Araujo, "From automation to augmentation: Human Resource's journey with Artificial Intelligence," *J. Ind. Inf. Integr.*, p. 100872, 2025.

[22] A. Montgomery, I. Todorova, A. Baban, and E. Panagopoulou, "Improving quality and safety in the hospital: The link between organizational culture, burnout, and quality of care," *Br. J. Health Psychol.*, vol. 18, no. 3, pp. 656–662, 2013.

[23] A. Fenwick, G. Molnar, and P. Frangos, "The critical role of HRM in AI-driven digital transformation: a paradigm shift to enable firms to move from AI implementation to human-centric adoption," *Discov. Artif. Intell.*, vol. 4, no. 1, p. 34, 2024.

[24] R. T. Sutton, D. Pincock, D. C. Baumgart, D. C. Sadowski, R. N. Fedorak, and K. I. Kroeker, "An overview of clinical decision support systems: benefits, risks, and strategies for success," *NPJ Digit. Med.*, vol. 3, no. 1, p. 17, 2020.

[25] M. A. Alabdali, S. A. Khan, M. Z. Yaqub, and M. A. Alshahrani, "Harnessing the power of algorithmic human resource management and human resource strategic decision-making for achieving organizational success: an empirical analysis," *Sustainability*, vol. 16, no. 11, p. 4854, 2024.

[26] A. Haleem, M. Javaid, R. P. Singh, and R. Suman, "Medical 4.0 technologies for healthcare: Features, capabilities, and applications," *Internet Things Cyber-Physical Syst.*, vol. 2, pp. 12–30, 2022.

[27] A. Lamberti-Castronuovo, M. Valente, F. Barone-Adesi, I. Hubloue, and L. Ragazzoni, "Primary health care disaster preparedness: a review of the literature

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content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2026.3688762

and the proposal of a new framework," *Int. J. Disaster Risk Reduct.*, vol. 81, p. 103278, 2022.

[28] M. Eshghali, D. Kannan, N. Salmanzadeh-Meydani, and A. M. Esmaieeli Sikaroudi, "Machine learning based integrated scheduling and rescheduling for elective and emergency patients in the operating theatre," *Ann. Oper. Res.*, vol. 332, no. 1, pp. 989–1012, 2024.

[29] R. R. Dangi, A. Sharma, and V. Vageriya, "Transforming healthcare in low-resource settings with artificial intelligence: Recent developments and outcomes," *Public Health Nurs.*, vol. 42, no. 2, pp. 1017–1030, 2025.

[30] G. Improta, V. Mazzella, D. Vecchione, S. Santini, and M. Triassi, "Fuzzy logic--based clinical decision support system for the evaluation of renal function in post-Transplant Patients," *J. Eval. Clin. Pract.*, vol. 26, no. 4, pp. 1224–1234, 2020.

[31] M. Shoaib *et al.*, "A deep learning-assisted visual attention mechanism for anomaly detection in videos," *Multimed. Tools Appl.*, vol. 83, no. 29, pp. 73363–73390, 2024.

[32] Y. Li, Y. Yao, J. Lin, and N. Wang, "A Deep Learning Algorithm Based on CNN-LSTM Framework for Predicting Cancer Drug Sales Volume," *arXiv Prepr. arXiv2506.21927*, 2025.

[33] N. Karunanayake, "Next-generation agentic AI for transforming healthcare," *Informatics Heal.*, vol. 2, no. 2, pp. 73–83, 2025.

[34] E. Y. Hidayat *et al.*, "Genetic algorithm-based convolutional neural network feature engineering for optimizing coronary heart disease prediction performance," *Healthc. Inform. Res.*, vol. 30, no. 3, pp. 234–243, 2024.

[35] T. Iqbal *et al.*, "Stress monitoring using wearable sensors: a pilot study and stress-predict dataset," *Sensors*, vol. 22, no. 21, p. 8135, 2022.

[36] J.-A. Sim, X. Huang, M. R. Horan, J. N. Baker, and I.-C. Huang, "Using natural language processing to analyze unstructured patient-reported outcomes data derived from electronic health records for cancer populations: a systematic review," *Expert Rev. Pharmacoecon. Outcomes Res.*, vol. 24, no. 4, pp. 467–475, 2024.

[37] A. S. Albahri *et al.*, "A systematic review of trustworthy artificial intelligence applications in natural disasters," *Comput. Electr. Eng.*, vol. 118, p. 109409, 2024.

[38] R. Kumar, A. Singh, A. S. A. Kassar, M. I. Humaida, S. Joshi, and M. Sharma, "Leveraging Artificial Intelligence to Achieve Sustainable Public Healthcare Services in Saudi Arabia: A Systematic Literature Review of Critical Success Factors.," *Comput. Model. Eng. Sci.*, vol. 142, no. 2, 2025.

[39] R. Liu *et al.*, "Enhanced detection classification via clustering svm for various robot collaboration task," in *2024 6th International Conference on Communications, Information System and Computer Engineering (CISCE)*, 2024, pp. 1121–1125.

[40] H. S. Ullah and S. Aslam, "Blockchain in healthcare and medicine: A contemporary research of applications, challenges, and future perspectives," *arXiv Prepr. arXiv2004.06795*, 2020.

[41] I. T. Akbasli, A. Z. Birbilen, and O. Teksam, "Artificial intelligence-driven forecasting and shift optimization for pediatric emergency department crowding," *JAMIA open*, vol. 8, no. 2, 2025.

[42] G. Arji, L. Erfannia, M. Hemmat, and others, "A systematic literature review and analysis of deep learning algorithms in mental disorders," *Informatics Med. unlocked*, vol. 40, p. 101284, 2023.

[43] S. M. Varkiani, F. Pattarin, T. Fabbri, and G. Fantoni, "Predicting employee attrition and explaining its determinants," *Expert Syst. Appl.*, vol. 272, p. 126575, 2025.

[44] S. G. Paul, A. Saha, M. Z. Hasan, S. R. H. Noori, and A. Moustafa, "A systematic review of graph neural network in healthcare-based applications: Recent advances, trends, and future directions," *IEEE Access*, vol. 12, pp. 15145–15170, 2024.

[45] Y. Sun and Y. Gao, "A multi-objective particle swarm optimization algorithm based on gaussian mutation and an improved learning strategy," *Mathematics*, vol. 7, no. 2, p. 148, 2019.

[46] S. Maleki Varnosfaderani and M. Forouzanfar, "The role of AI in hospitals and clinics: transforming healthcare in the 21st century," *Bioengineering*, vol. 11, no. 4, p. 337, 2024.

[47] X. Wang, H. Yu, S. Kold, O. Rahbek, and S. Bai, "Wearable sensors for activity monitoring and motion control: A review," *Biomim. Intell. Robot.*, vol. 3, no. 1, p. 100089, 2023.

[48] N. Savanović *et al.*, "Intrusion detection in healthcare 4.0 internet of things systems via metaheuristics optimized machine learning," *Sustainability*, vol. 15, no. 16, p. 12563, 2023.

[49] S. Golubovic, L. Jovanovic, B. Radomirovic, A. Njegus, M. Zivkovic, and N. Bacanin, "Evolving deep neural network architectures by sine cosine algorithm for healthcare 4.0," in *2023 IEEE International Conference on Contemporary Computing and Communications (InC4)*, 2023, vol. 1, pp. 1–6.

[50] F. Markovic *et al.*, "Parkinsons detection from gait time series classification using modified metaheuristic optimized long short term memory," *Neural Process. Lett.*, vol. 57, no. 1, p. 14, 2025.

[51] L. Jovanovic, N. Bacanin, M. Zivkovic, M. Antonijevic, A. Petrovic, and T. Zivkovic, "Anomaly detection in ecg using recurrent networks optimized by modified metaheuristic algorithm," in *2023 31st telecommunications forum (TELFOR)*, 2023, pp. 1–4.

[52] L. Jovanovic, S. Golubovic, N. Bacanin, G. Kunjadic,

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M. Antonijevic, and M. Zivkovic, "Performance evaluation of metaheuristics-tuned deep neural networks for healthcare 4.0," in *International Conference on Computational Sciences and Sustainable Technologies*, 2023, pp. 1–14.

[53] S. J. Villoth *et al.*, "Cardiovascular Disease Risk Prediction Utilizing Two-Tier Classification Framework Optimized with Adapted Variable Neighborhood Search Algorithm," *Algorithms*, vol. 19, no. 2, 2026.

[54] L. Anicin *et al.*, "Metaheuristic-driven dual-layer model for classifying Alzheimer's disease stages," *Front. Comput. Neurosci.*, vol. Volume 20-2026, 2026.

[55] P. Dabic *et al.*, "Hospital Admission Classification of Cardiac Patients Utilizing Metaheuristics-Optimized Two Tier Framework," no. 2026, 2025.

[56] N. R. Hander *et al.*, "Working conditions, job stress and work-related consequences among hospital employees-differences by professional group, working hours and job levels: A cross-sectional study.," *PLoS One*, vol. 21, no. 3, p. e0343567, 2026.

[57] J. A. Zafra-Agea, E. Ramírez-Baraldes, E. Maldonado-Manzano, N. Obradors-Rial, A. Puiggrós-Binefa, and E. Colillas-Malet, "Working Conditions and Well-Being of School Nurses in Spain: Impact on Job Satisfaction and Professional Quality of Life," *Healthcare*, vol. 13, no. 3, 2025.

[58] M. Sánchez-sellero, "SYNTHE TIC INDIC ATORS OF QUALIT Y OF SUBJEC TIVE LIFE IN THE EU : RUR AL AND URBAN ARE AS," *Prague Econ. Pap.*, vol. 30, no. 5, pp. 529–551, 2021.

[59] Z. Zhang and M. Tao, "Deep Learning for Wireless Coded Caching with Unknown and Time-Variant Content Popularity," *IEEE Trans. Wirel. Commun.*, vol. 20, no. 2, pp. 1152–1163, 2021.



MUHAMMAD SHOAIB is an accomplished researcher with expertise in Medical Image and Signal Processing, Machine Learning, computer vision, image understanding, pattern recognition, video scene understanding, and Human activity analysis. He obtained his Master's degree in Computer Science from the Department of Computer Science, Islamia College Peshawar, Khyber Pakhtunkhwa, Pakistan, and is currently pursuing his Ph.D. degree in Computer Science from Sarhad University of Science and Technology Peshawar, Khyber Pakhtunkhwa, Pakistan. With his extensive knowledge and expertise, Muhammad Shoaib has contributed significantly to the field of computer science, having published numerous research articles in reputed academic journals. His work has been well-received by the academic community and has helped to advance our understanding of these complex topics. Currently, he is serving as a Lecturer at the Department of Computer Science, CECOS University of IT and Emerging Sciences Peshawar, Khyber Pakhtunkhwa, Pakistan, where he is actively engaged in teaching and research activities.

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MUHAMMAD ISMAIL MOHMAND is currently working as an Associate Professor in the Department of Computer Engineering, Faculty of Engineering and Natural Sciences, at Istanbul Atlas University, Turkey. He has published original research articles in various international and reputed journals, and he is actively engaged in both teaching and research. His research interests include Computer Networking, Artificial Intelligence, Cybersecurity, the Internet of Things (IoT), and Software Engineering.



NASIR SAYED received the Master's degree in Computer Science from the Department of Computer Science, Islamia College Peshawar, Khyber Pakhtunkhwa, Pakistan, where he is currently pursuing the Ph.D. degree in Computer Science. His research interests focus on intelligent computer vision–based systems, including human activity recognition, video analysis, deepfake detection, and medical image analysis. His current research involves the application of deep learning and machine learning techniques for feature representation, pattern recognition, and robust decision-support systems in real-world and healthcare-oriented environments.



INAYAT KHAN received a PhD degree in Computer Science from the Department of Computer Science, University of Peshawar, Pakistan. He is currently working as an Assistant Professor in the Department of Computer Science, University of Engineering and Technology Mardan, Pakistan. His current research is based on the design and development of context-aware adaptive user interfaces for minimizing drivers' distractions. His other research interests include lifelogging, Healthcare, Deep learning, ubiquitous computing, accessibility, Mobile-based assistive systems for people with special needs, and published several articles in international journals and conferences in these areas.



SALMAN JAN received a master's degree in computer science from the University of Peshawar and the PhD degree from the MIIT, Universiti Kuala Lumpur, in 2019. His PhD thesis was on Android malware analysis and its integration with blockchain for preserving integrity of the behavioural logs and ultimately to secure the classification results of the behavioral logs through the employed Deep Learning models including DCGAN, CNN, FCNN. He has acquired skills and published in a number of well-reputed journals in areas of machine learning, deep learning, artificial intelligence, blockchain, IoT, and security Augmented and Virtual Reality. He also is actively working on international projects in the capacity of consultant and successfully completed numerous international projects. He is presently serving as Assistant Professor at Faculty of Computer Studies, Arab Open University, Kingdom of Bahrain.



IT EE LEE received the Bachelor of Engineering degree (with Hons.) in electronics and the M.Eng.Sc. degree from Multimedia University, Cyberjaya, Malaysia, in 2003 and 2009, respectively, and the Ph.D. degree specializing in optical wireless communication from the University of Northumbria, Newcastle upon Tyne, U.K., in 2014. She is currently an Assistant Professor with the Faculty of Artificial Intelligence & Engineering, Multimedia University, Cyberjaya, Malaysia. She has authored numerous publications in international peer-reviewed journals, international conferences, and book chapter. Her research interests include wireless optical communications, Internet of Things, machine learning, and sustainable technologies. She also contributes as a technical program committee member for international conferences and a technical reviewer for various distinguished international peer-reviewed journals.

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